

Master's Programme in Information and Service Management

A Service Operations Perspective on Educational Efficiency: Benchmarking Aalto University School of Business

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Abstract

This thesis investigates the teaching efficiency of Aalto University School of Business (Aalto BIZ) relative to selected European business schools and other universities. It is motivated by the need to understand how Aalto BIZ's resource use translates into measurable educational outputs under increasing pressure for public-sector accountability and efficient resource use. The thesis applies a service operations perspective to higher education, arguing that teaching is a service defined by its extensive customer participation and dependence. This creates challenges for defining outputs, separating student and institutional contribution, and defining quality.

The empirical analysis uses data envelopment analysis (DEA) to benchmark relative teaching efficiency with data from the European Tertiary Education register (ETER), primarily averaged over the period 2015-2021 to ensure comparability of benchmarked institutions. Institutions are compared in their ability to turn resources into enrolled students and enrolled students into graduates, and they are grouped into clusters to account for differences in their composition and improve comparability. The results suggest that Aalto BIZ is highly efficient compared to other universities but around average among sampled business schools. Its efficiency is driven primarily by high student counts relative to resource use. However, its longer than average graduation times, particularly in master's programs, reduce institutional efficiency.

The thesis contributes to service operations literature by introducing a theoretical framework on service operations classification. It also extends the findings from service operations to higher education benchmarking by proposing a two-stage model treating enrolled students as intermediate products, participating in the production of graduates. The findings suggest that there are differences in institutional efficiency based on disciplinary composition, environment, and strategy in addition to managerial efficacy.

Keywords data envelopment analysis, operations management, university efficiency, service operations management, service theory, productivity

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Tiivistelmä

Tässä tutkielmassa arvioidaan Aalto-yliopiston kauppakorkeakoulun (Aalto BIZ) opetuksen tehokkuutta suhteessa valikoituihin eurooppalaisiin kauppakorkeakouluihin ja muihin yliopistoihin. Tutkimuksen tavoitteena on selvittää, kuinka tehokkaasti Aalto BIZ muuntaa käytettävissään olevat resurssit koulutuksen tuotoksiksi. Tutkimusta motivoi lisääntynyt kiinnostus julkisten palveluiden kuten korkeakoulujen tilivelvollisuutta ja resurssien käyttöä kohtaan.

Tutkielmassa koulutusta käsitellään palveluoperaatioteorian kautta. Opetus nähdään palveluna, jonka tuotanto perustuu opiskelijoiden laajaan osallistamiseen ja heidän panokseensa tuotantoprosessissa. Tämä vaikeuttaa koulutuksen tuotosten ja laadun määrittelyä sekä opiskelijoiden ja korkeakoulujen vaikutusten erottamista toisistaan.

Empiirisessä analyysissä opetuksen suhteellista tehokkuutta arvioidaan data envelopment analysis (DEA) -menetelmällä käyttäen European Tertiary Higher Education Research (ETER) -aineistoa. Analyysi keskittyy pääosin keskiarvotettuun vuosien 2015–2021 aineistoon, jotta aineisto olisi mahdollisimman vertailukelpoista. Tehokkuutta tarkastellaan kaksivaiheisella DEA-mallilla, joka mittaa resurssien muuttamista opiskelijoiksi ja edelleen opiskelijoiden muuttamista valmistuneiksi. Vertailtavat korkeakoulut jaetaan klustereihin, jotta niiden rakenteelliset erot huomioidaan.

Tulosten perusteella Aalto BIZ on erittäin tehokas verrattuna muihin yliopistoihin mutta sen tehokkuus on keskiarvoista muihin kauppakorkeakouluihin nähden. Sen vahvuus on suuri opiskelijamäärä suhteessa käytettyihin resursseihin, kun taas keskimääräistä pidemmät valmistumisajat heikentävät koulun kokonaistehokkuutta.

Tutkielma esittää palveluoperaatioiden luokittelua kuvaavan teoreettisen viitekehyksen. Lisäksi kaksivaiheinen malli laajentaa palveluoperaatioiden näkökulman soveltamista korkeakoulutuksen tehokkuuden arviointiin. Tulokset osoittavat, että johdon lisäksi korkeakoulujen tehokkuuteen vaikuttavat strategia, tieteenalakoostumus ja toimintaympäristö.

Avainsanat data envelopment analysis, operaatiotutkimus, palveluoperaatiojohtaminen, palveluteoria, tuottavuus, yliopistojen tehokkuus

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Abbreviations and glossary

Abbreviations

Aalto BIZ	Aalto University School of Business
DEA	Data envelopment analysis
DMU	Decision-making unit
FTE	Full-time equivalent
HEI	Higher education institution
PPP	Purchasing power parity

Glossary

Benchmarking	Comparison of an institution's performance against comparable peers to identify relative strengths and improvement potential
Capacity efficiency (stage 1)	Efficiency with which institutional resources are converted into enrolled students
Completion efficiency (stage 2)	Efficiency with which enrolled student population is converted into graduates
Efficiency	Relative ability of an institution to transform inputs into outputs compared with best-performing peers
Frontier	The efficient production boundary formed by the best performing institutions in the sample
Learning	The process through which students acquire knowledge, skills, and competencies
Output	The direct products of the educational production process, such as graduates
Outcome	The broader effects of higher education beyond direct output such as graduate employability
Production technology	The effective input-output transformation process (i.e. the production function) of an organisation
Productivity	Performance measured independently of peer comparisons, typically reflecting change in outputs relative to inputs over time
Teaching	Institutional activities that provide instruction, supervision, and learning opportunities to students

1 Introduction

In recent decades, efficiency has become increasingly important in the public sector, as institutions are expected to not only provide social value but demonstrate efficient resource use. Higher education is a central example as governments including Finland seek to raise educational attainment without large commitments to increase public funding. Between 2015 and 2022, real expenditure per student in Finnish tertiary education fell by 14% (OECD, 2025b). This development has made efficiency measurement an increasingly important concern in higher education policy and management (Johnes et al., 2017; Jongbloed et al., 2018; Massy et al., 2013). This thesis is a part of Aalto University School of Business' (Aalto BIZ) project to improve understanding on the school's resource use efficiency.

This thesis primarily evaluates relative teaching efficiency: how effectively institutions transform resources into educational outputs compared with best-performing comparable peers. In contrast, productivity describes the relationship between inputs and outputs irrespective of comparison with other institutions. Universities produce multiple outputs, many of which are delayed, intangible, or only partially attributable to the institution (Belfield & Fielding, 2001; Hanushek & Ettema, 2017; Johnes, 2006). Inputs are similarly difficult to quantify as varying environments and students contribute to observed outcomes (Hanushek, 1979; Parteka & Wolszczak-Derlacz, 2013). These characteristics make productivity difficult to summarise using a single input-output ratio. Accordingly, this thesis uses efficiency for benchmarking, while productivity is considered briefly when analysing changes in Aalto BIZ's performance over time using the Malmquist productivity index and descriptive input-output ratios.

Because the number of suitable domestic benchmarking institutions is limited, a cross-country sample was preferred. The Bologna declaration of 1999 initiated a process of harmonisation across European higher education. Combined with the existence of the European Tertiary Education Register (ETER), which provides strictly validated data to promote cross-European comparability (Lepori et al., 2016), building a sample of European universities was deemed most suitable. These data are analysed through data envelopment analysis (DEA) with business schools as the primary focus of benchmarking, and clustering is employed to increase the number of benchmarks while preserving comparability.

While much of the previous literature employs DEA to examine broad efficiency patterns across higher education systems (De Witte & López-Torres, 2017), this thesis applies DEA as a benchmarking tool to evaluate the teaching efficiency of Aalto BIZ against comparable European institutions.

Particular emphasis is placed on how to define the higher education teaching production process and how to feasibly model it with available data in a robust way to provide information for institutional decision-making.

This thesis approaches the problem of educational productivity from a service operations perspective, particularly through the work of Chase (1978), Frei (2006), Grönroos (1984), and Sampson & Froehle (2006). Unlike services such as equipment maintenance where the provider can perform much of the production independently of the customer, educational production requires continuous customer participation (Sampson & Froehle, 2006). Students participate in the production process as suppliers (Sampson, 2000): they provide effort, prior knowledge, motivation, time, and choices about study pace and specialisation (Hanushek, 1979; Johnes et al., 2017). This distinction matters for efficiency measurement because observed teaching outputs reflect both institutional resources and student-provided inputs (Hanushek, 1979; Parteka & Wolszczak-Derlacz, 2013). Existing studies typically represent teaching as a single production process linking institutional resources directly to educational outputs (De Witte & López-Torres, 2017). This thesis uses a two-stage model to distinguish between capacity and completion efficiency and recognise students as the output of institutional resource use but also a key input into completion. It also considers the production process of higher education teaching to inform further research in developing efficiency measurement at Aalto BIZ.

The research questions are set out as follows:

RQ1: What is the relative teaching efficiency of Aalto BIZ compared to its peers?

RQ2: How has teaching efficiency at Aalto BIZ developed over time, and how does this development compare to its peers?

The following thesis is divided into four chapters. The first chapter has introduced efficiency measurement and its relevance in higher education. Chapter 2 builds a theoretical framework for classifying services and demonstrates the challenges in higher education teaching efficiency measurement and interpretation. It also reviews previous literature on how to benchmark efficiency in the presence of these limitations. Chapter 3 outlines the methodology and the data as well as the robustness of the research. In the fourth chapter, results are presented and analysed. Chapter 5 discusses the theoretical and managerial implications of the results and considers its limitations.

2 Literature review

This literature review develops the theoretical and methodological basis for benchmarking higher education teaching productivity. It first uses service operations theory to explain why higher education differs from more standardised production settings (Sampson, 2000; Sampson & Froehle, 2006). Students are not merely recipients of outputs but also provide key inputs and participate in production which creates challenges for operational efficiency, quality evaluation, and productivity measurement (Hanushek, 1979; Sampson, 2000). The review then examines how productivity and efficiency have been measured in higher education, why frontier methods are suitable in this context, and what prior studies imply for interpreting efficiency estimates (De Witte & López-Torres, 2017; Worthington, 2001). The chapter concludes with a literature framework for how to approach higher education teaching productivity measurement and interpret the results.

2.1 Types of service operations

Developed economies have undergone a transition from goods production to services, which now constitute the dominant share of economic activity (Triplett & Bosworth, 2004; WTO, 2023). In Finland, services accounted for approximately 70% of value added in 2023, closely matching the OECD average (OECD, 2025b). It has become increasingly important to consider how to define service operations and the ways their operational challenges and measurement differ from other sectors such as manufacturing. This section considers how to define service operations as distinct from other production and how the customer's role influences their efficiency potential and quality formation.

The motivation for treating services as distinct from resource extraction and goods production lies in the need to understand their production constraints and managerial challenges that are not adequately captured by manufacturing-based models (Frei, 2006; Sampson, 2012; Silvestro et al., 1992). Scholars have proposed multiple criteria to differentiate services from goods production (Hill, 1977; C. Lovelock & Gummesson, 2004). When services accounted for a relatively small share of the economy, they were often considered a residual in three-sector models, remaining after subtracting primary and secondary activities of manufacturing and resource extraction (Clark, 1940; Fisher, 1939). This residual treatment of services became increasingly untenable as they grew into the dominant component of developed economies: by 2015, they accounted for about 70 percent of value added of high-income economies according to World Bank data. As a result, the need to develop a distinct, positively defined category became apparent (Hill, 1977). Early definitions focused on the characteristics of service outputs,

particularly intangibility, heterogeneity, inseparability and perishability, as summarised in the IHIP framework (Zeithaml et al., 1985). However, as the service sector expanded, this became an increasingly inadequate framing due to the proliferation of heterogeneous processes and hybrid offerings, bundling goods and services (C. Lovelock & Gummesson, 2004).

In contrast to IHIP, later frameworks moved away from output characteristics to the process through which value is created. Service-dominant logic was influential in this shift, as Vargo & Lusch (2004) argued that value is not simply embedded in outputs and delivered to passive consumers but co-created through interaction between provider and customer. However, as service-dominant logic generalises this reasoning to all exchange, it fails to provide a clear boundary between goods and services.

A more useful framework, and the view adopted by this thesis is Sampson & Froehle's (2006) unified service theory (UST), which offers a more precise process-based distinction by identifying customer-provided inputs as the distinguishing characteristic of services. In this view, customers contribute not only needs and expectations, but also necessary production inputs through their capabilities and participation. Sampson & Froehle reference Wemmerlöv (1990) who divides customer inputs into three broad categories: the customer's self, their tangible belongings, and information. More concretely, common customer inputs can include physical presence, skills, and knowledge or decision-making (Stewart & Chase, 1999).

2.1.1 Customer participation's impact on variation

Variation within service production processes is analytically central because the production technologies of service processes differ substantially across service types, in some cases more than the difference between highly standardised services and manufacturing (Silvestro et al., 1992). In line with DEA literature, production technology in this thesis refers to the effective input-output transformation structure of the production process: the relevant inputs, how they are combined, and how outputs are generated. In manufacturing, this relationship is often stable and provider controlled. In services, by contrast, this transformation is also shaped by customer-provided inputs, which are heterogeneous and only partially controllable. UST is useful as it defines services through this process-level dependence on customer inputs. However, as Spring & Araujo (2009) note, this remains only a starting point because UST does not account for the substantial variation within service production.

A general principle in operations management is that variation, where possible, should be reduced as it tends to reduce operational performance by

degrading efficiency, predictability, and consistency (Hopp & Spearman, 2021). Due to the customer's participation in the production process, the scope for this is more limited in some services (Frei, 2006; Sampson & Froehle, 2006). The interaction with customers introduces new challenges for producers as customer-provided inputs are heterogeneous, outside managerial control, and may generate unforeseen errors (Stewart & Chase, 1999). Consequently, service processes should be designed to handle additional contingencies and have less potential for standardisation. Frei (2006) argues that as firms cannot effectively eliminate customer-induced variation, they must determine which forms of variation to reduce and which to accommodate. Building effective processes in accordance with the chosen strategy can increase efficiency while also preserving or enhancing service quality (Frei, 2006; Hopp & Spearman, 2021).

To increase efficiency while preserving flexibility, many service producers adopt a mixed strategy where they standardise some parts of their production process while preserving discretion in others (Frei, 2006). Lillrank & Liukko (2004) provide a distinction between standard, routine and non-routine processes suggesting that differences in repetition, variety, and uncertainty determine standardisation potential. Service processes can therefore be decomposed into components to standardise routine processes and modularise service elements to maintain flexibility where customer interaction is more consequential (Pine et al., 1993; Silvestro et al., 1992). This way front-office and back-office work can be separated so that customer interaction occurs only when it contributes to value creation, while routine activities are performed in a controlled environment supporting standardisation and efficiency (Chase, 1978). By limiting discretion to stages where customer heterogeneity is most consequential, service firms can partially mitigate the efficiency losses associated with high customer participation while preserving perceived service quality (Metters & Vargas, 2000).

The iterative and individualised nature of service production means that quality in services is more complex than manufacturing as value is partly endogenous to the production process rather than embedded in the final product (Grönroos, 1984; Zeithaml, 1988). When products cannot be made to spec, or human interaction is a significant part of the customer experience, perceived quality is increasingly dependent on factors beyond the quality of the output.

2.1.2 Customer participation's impact on quality

Grönroos (1984) distinguishes between technical quality, referring to the service outcome received by the customer and functional quality, referring to how the service is delivered and experienced during the production process.

This distinction is particularly important in services such as education where customers provide significant inputs or participate directly in the production process.

With increasing customer participation and individualisation, appraising technical quality becomes more difficult as outputs are less comparable relative to standardised services (Zeithaml, 1981). At the same time, greater participation increases the customer's exposure to the production process, increasing its influence in customer evaluation of service quality (Grönroos, 1984). The importance of functional quality therefore tends to increase as the customer's role increases (Grönroos, 1984; Zeithaml, 1981). Although customisation may reduce standardisation, throughput, and predictability, it may improve customer fit therefore increasing perceived service quality and, in some cases, the quality of the outcome (Frei, 2006; Morris & Johnston, 1987). Quality evaluation and interaction of efficiency and quality in higher education are discussed more in chapter 2.2.2.

To illustrate service management differences between firms, Chase (1978, 2010) divides service processes according to the amount of customer contact, roughly defined as the percentage of time the customer must be in the system relative to the time it takes to serve them. However, customer contact time alone is not sufficient, as the production possibilities and therefore efficiency of service operations are also determined by how customers are involved in production and value creation (Sampson & Chase, 2022). These dimensions shape the extent to which production can be standardised, the level of discretion required by providers, and the degree to which performance depends on customer behaviour and input quality (C. H. Lovelock, 1983; Silvestro et al., 1992). In addition, the role of the customer also affects the relative perception and importance of technical and functional quality. Therefore, service operations with differing modes of customer involvement will vary in efficiency potential and quality formation.

2.1.3 Service categorisation based on customer participation

The preceding discussion shows that service operations differ not merely by the degree of contact, but by the role customers play as productive inputs, sources of variation, and co-creators of value. A further analysis of service operations requires attention to the different modes through which customers participate in production. To determine efficiency and quality implications of customer involvement, we need to consider the significance of and reliance on customer contributions rather than pure contact time. Considering the customer's role in accordance with Sampson & Froehle (2006), a classification is presented in Figure 1 based on primary customer participation mode in production.

This taxonomy is derived by synthesising existing process-based views of service. UST takes a customer-centric view and defines services through customer-provided inputs (Sampson & Froehle, 2006), while operations literature takes a process centric view based on customer contact and process standardisation (Chase, 1978; Silvestro et al., 1992). Combining these two views, we can synthesise how process and customer demands intersect to create a taxonomy of service operations based on customer participation mode. Services on the left of the continuum have processes where customer participation can remain bounded while services on the right are more reliant on customers for key processes. The extent of these inputs has a significant effect on service processes and therefore affects standardisation and efficiency potential as well as quality evaluation.

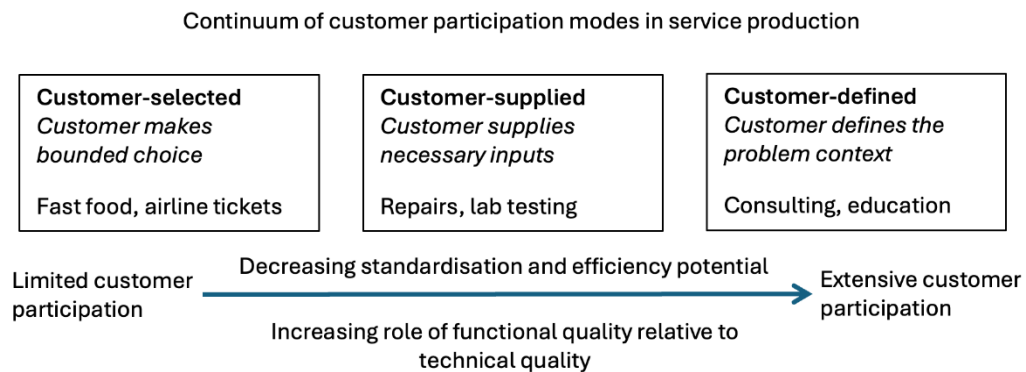


Figure 1. Service operations taxonomy according to primary customer participation mode.

Production processes should be categorised according to their highest mode of customer participation they require since the existence of higher modes simultaneously imply the existence of lower ones. The mode may also depend on the customer problem within a field, for example, routine healthcare falls under input provision as many diagnoses are standard while in more complex cases, the patient requires more individualised and novel long-term ongoing diagnosis and treatment, implying a longer co-defined process.

A way to categorise services is to consider where the customer and provider's value chains intersect. When the customer only participates in selection, the provider can produce a standardised offering with only a few alternatives, such as airline tickets where routes are planned in advance and customers book standard offerings choosing between different tiers of amenities attached to the product. Services where this is possible, can resemble manufacturing in many of their processes. When customers supply significant production critical inputs, such as in repair services, the production cannot be

started without the customer and thus may vary depending on the input. When customers have complex and more open-ended problems, their participation is required throughout the process, and the full extent of the service product or output is often not fully specified at the point of first contact between customer and provider.

In summary, the key distinction adopted here separating service production from manufacturing is the involvement of the customer in the production process in accordance with Sampson & Froehle (2006). Their customer centric view is adapted here to service classification, where customer participation influences both efficiency potential and quality evaluation extending the works of Chase (1978) and Grönroos (1984). As customer participation becomes earlier, more intensive, and less standardised the provider's control of the production process is reduced. Firms may respond by limiting or managing this variability, leading to potential trade-offs in customer experience and therefore subjective quality and service efficiency. Particularly in high participation services this variability is a key component of value to provide solutions for highly complex, individual problems. These services often provide a lot of value to customers, but they are also complex and costly to produce. Table 1 outlines some of the relevant characteristics of services with different levels of customer participation. This complements Figure 1 by showing the practical implications of the service's position on the customer participation continuum.

Table 1. Dimensions of differences between service production processes according to primary customer participation mode.

Dimension	Customer-selected	Customer-supplied	Customer-defined	Theoretical basis
Customer role	Chooser	Input provider	Co-producer	(Chase, 1978; Sampson & Froehle, 2006)
Process type	Standard	Routine	Non-routine	(Lillrank & Liukko, 2004)
Variation source	Demand mix	Input heterogeneity	Joint interaction	(Frei, 2006; Sampson & Froehle, 2006)
Efficiency potential	High via standardisation	Conditional on managing input variability	Constrained by interaction demands	(Sampson, 2018)
Quality evaluation	Technical weighted	Mixed technical and functional	Functional weighted	(Grönroos, 1984)

2.2 A service operations perspective on higher education

This section applies the preceding service operations framework from Figure 1, particularly the concepts of customer-provided inputs, customer-induced variation, and customer participation in production, to higher education teaching (Chase, 1978; Grönroos, 1984; Sampson & Froehle, 2006). Within the taxonomy presented in Figure 1, higher education teaching is located in the problem-definition category because in addition to selecting an offering and providing initial inputs, students also shape their study paths and influence the eventual output through their contribution throughout production. Section 2.2.1 examines the higher education production process and its relevant inputs and outputs summarised in Figure 2, while section 2.2.2 considers how quality is defined and how it could be measured.

2.2.1 Higher education production process and measurement

This section develops a framework of the higher education teaching production process depicted in Figure 2 to evaluate the relevant variables for efficiency benchmarking. Higher education is often classified as a part of professional services, which are characterised by high levels of customer contact and employ highly trained specialists performing unstructured, complex tasks (Sampson, 2018; Silvestro et al., 1992). This makes them very different from manufacturing but also distinguishes them from some of the more standardised services, lower customer participation services, such as postal services and retail banking, sometimes labelled quasi-manufacturing (Chase, 2010, pp. 11–17). Higher education institutions (HEIs) are complex service providers with a multi-input, multi-output production function (Cossani et al., 2022; Parteka & Wolszczak-Derlacz, 2013). In addition, multiple parties can be considered stakeholders including students, governments, and employers (Doherty, 2008; Jongbloed et al., 2018). From the standpoint of the production process, students are the active participants and therefore considered the higher education customers in this thesis (Chase, 1978; Sampson & Froehle, 2006).

Higher education is not straightforward to position on the continuum presented in Figure 1 because it combines formal standardisation with intensive customer participation. Degree structures, curricula, and assessment requirements define the broad boundaries of the process, but they do not determine the learning that occurs within them. Knowledge cannot simply be transferred to a passive recipient, and learning outcomes depend partly on student-provided inputs such as prior preparation, ability, effort, motivation, and continued engagement (Goldhaber & Brewer, 1997; Hanushek, 1979). The process is also cumulative and interactive: later learning depends on

capabilities developed earlier and students require varying levels of support, meaning a rigid, standard process is unlikely to serve all students well (Dochy et al., 1999; Kalyuga et al., 2003). Students shape the production process through choices concerning specialisation, course selection, study pace, depth of engagement, and intended academic or labour-market outcomes. Higher education students are not merely recipients of a predefined service, but both a critical input and an active participant in the production process (Robledo, 2024). Although the formal credential is specified in advance, neither the detailed production process nor the resulting development in the student can be fully determined at the outset.

Customer participation in higher education is distinctive because, in addition to students defining the goals and constraints, they are the objects of transformation, and their learning its principal output. Similar process characteristics exist in coaching and development-oriented services where the provider can structure and support the process, but the output is non-separable change in the customer. Examples include psychotherapy, structured skill development such as musical instruction, and physiotherapy. These processes differ in how separable and measurable their outputs are. HEIs provide teaching whose output is student learning: the development of knowledge, skills, and capabilities that exist within the student and depend on the student's attributes and continued participation (Robledo, 2024). Teaching is the main institutional contribution to the production process, whereas learning is the desired output, and is partly endogenous to the student.

Sampson (2000) uses the concept of customer-supplier duality to describe service processes in which the customers both receive outputs from the provider and supply essential inputs into production. The higher education production process is defined by co-production as student-provided inputs and institutional resources converge to create outputs. The nature of the production process is cumulative and recursive as students build on prior skills and design their study paths within institutional constraints.

Because human capital is largely latent, it cannot be measured consistently at entry and exit (Hanushek & Ettema, 2017). Therefore, value-added approaches suffer from difficulty in defining inputs and outputs, selection effects, and poor comparability between different nations, particularly in public and non-market sectors (Angrist et al., 2017; Hanushek, 1979; Hanushek & Ettema, 2017; Johnes et al., 2017). The variation in student characteristics is a key cause of observed performance differences across institutions, complicating the attribution of outcomes to institutional effectiveness (Goldhaber & Brewer, 1997; Hanushek, 1979). Therefore, the productivity of education is measured through proxies to approximate the transformation.

Degrees are a common proxy and represent the most standardised output of universities, but they too have differing value between fields, institutions, and students (Belfield & Fielding, 2001; Marginson, 2008). They are administratively standardised but stand in for a heterogeneous set of educational outcomes, representing the increase in the student's human capital. Many outcomes of higher education emerge over time and are partly dependent on students' participation (Belfield & Fielding, 2001). While broader outcomes such as labour market performance, knowledge transfer, and learning gains are sometimes used, they are less suitable as direct production outputs due to time lags, endogeneity, and measure inconsistency. Degrees remain a practical output indicator, especially when comparison is limited to a relatively homogenous group of institutions (Herberholz & Wigger, 2021; Johnes, 2006).

Even in the case of degrees, the production occurs over multiple years, with substantial variation in time-to-degree across students and institutional contexts influenced by their environment (OECD, 2025). This creates a mismatch between the timing of inputs and observed outputs, as resources deployed in a given period may not correspond to completions observed within the same timeframe. Such temporal misalignment introduces additional measurement error and weakens the interpretation of contemporaneous productivity estimates (G. Johnes & Johnes, 2009). This is particularly relevant when institutional scale changes, as inputs must be increased years before most of the measured output gains are realised. Consequently, degrees are a practical proxy for educational output, but meaningful comparisons require managing heterogeneity and aligning inputs to outputs through lagged output variables.

The other key *output* of universities is research. Research outputs are more directly observable than teaching outputs, but are still measured through proxies such as publication counts, citations, and publications in high-level journals. While these indicators capture quantity and perceived quality, they vary across fields and are influenced by publication practices (Hicks et al., 2015; Moed, 2005; Waltman, 2016). Teaching and research are jointly produced with a shared pool of institutional inputs, particularly faculty time, which creates allocation trade-offs between the two activities (De Witte & López-Torres, 2017; Johnes, 2006). At the same time complementarities may arise, as research activity can enhance teaching through the integration of current knowledge and expertise, although empirical evidence for this is mixed (Hattie & Marsh, 1996; Marsh & Hattie, 2002).

Combined with the *quality* focus in research evaluation and the highly skewed distribution of journal placement and citation (Clauzet et al., 2009; Merton, 1968), this leads to output measures which are driven

disproportionately by few high-performing individuals. Therefore, research output does not simply reflect institutional management quality, but also the composition of researchers, making it sensitive to selection effects (Stephan, 2015). Measuring efficiency by combining research and teaching output may therefore distort results due to differences in resource allocation and individual-level variation rather than institutional performance. Simultaneous measurement also creates an identification problem as the allocation of shared resources between teaching and research is rarely observed, outputs cannot be cleanly attributed to resource use.

Higher education *inputs* can be divided into institutional, student-provided, and environmental (Johnes, 2006; Parteka & Wolszczak-Derlacz, 2013). Institutional inputs include teaching staff, administration, and physical and digital infrastructure, typically measured by personnel counts or expenditure. Student-provided inputs, such as prior ability, motivation, and engagement are critical to production but difficult to observe, while environmental variables such as funding models or labour market structures shape the behaviour of institutions and student as well as observed performance without direct institutional control (Marginson, 2008; OECD, 2025).

Previously students have been discussed in both inputs and outputs, and their role is a recurring question in efficiency research (Worthington & Lee, 2008). In models that measure access to educational resources, students may be treated as an output but from a production perspective focused on learning transformation, students function as inputs and intermediate products (Sampson & Froehle, 2006). Institutional resources first create educational capacity by attracting and supporting an enrolled student body. That enrolled stock then becomes a key input into degree completion through continued student participation. This interpretation follows directly from the service co-production structure, where students act in the dual role as customers and suppliers (Sampson, 2000), and in education are also objects of transformation through production. This customer-supplier-object triad extends the previous literature and provides the theoretical basis for the two-stage modelling approach outlined in the methodology.

Educational *outcomes* emerge through co-production where institutional and student resources interact. This relationship is reinforced by peer effects as students contribute to their individual outcomes and shape the learning environment faced by others (Johnes et al., 2017; Winston & Zimmerman, 2004). A direct result is that observed outputs of higher education reflect both selection and transformation. Institutions select students from heterogeneous applicant pools and outcomes depend on prior ability and future engagement. At the same time institutions can adjust teaching practices, assessment, and student support to the characteristics of their student cohorts,

making student composition partly endogenous to managerial and pedagogical decisions (Star & McDonald, 2007). Figure 2 shows a simplified version of the higher education teaching production process.

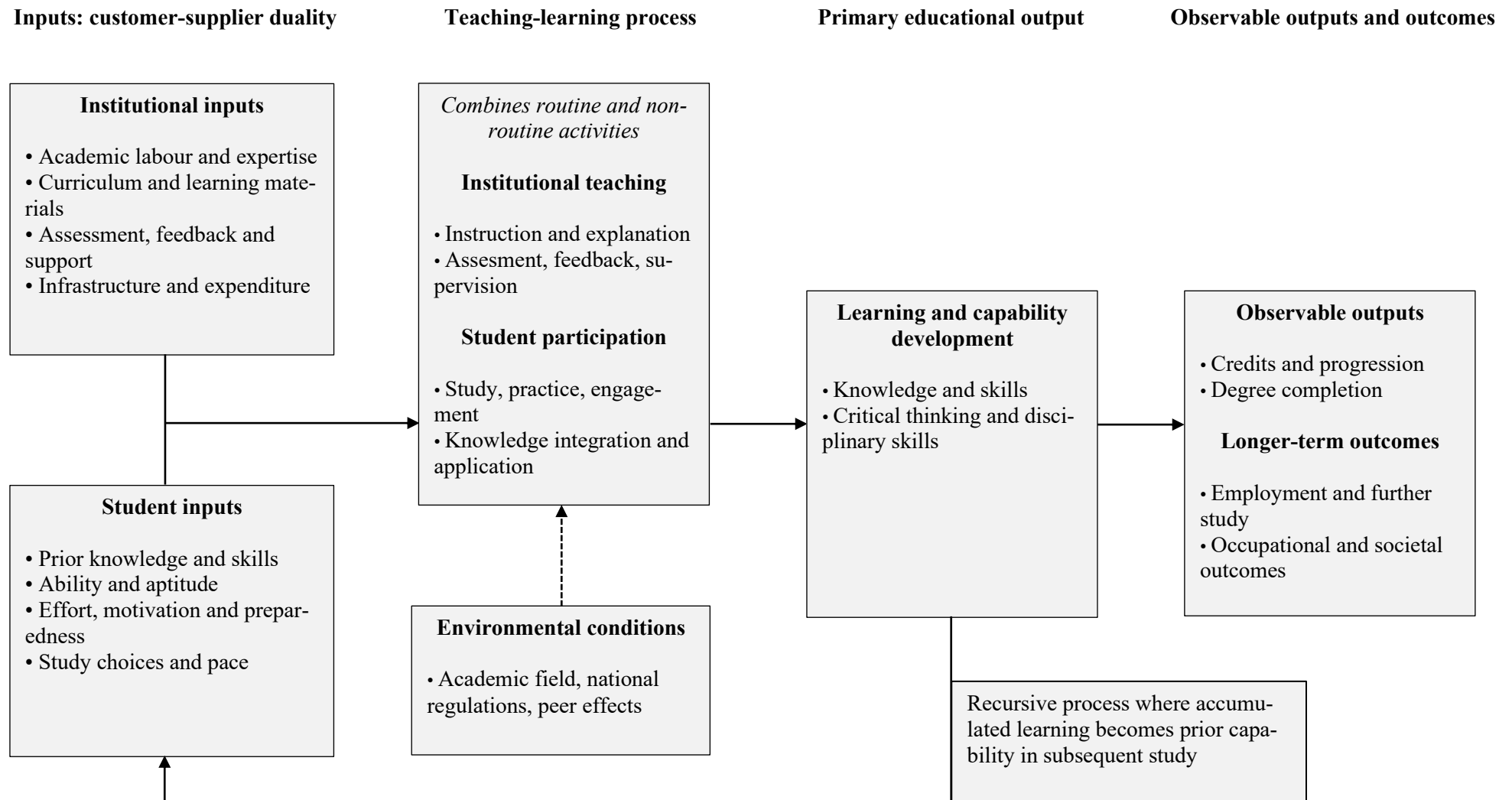


Figure 2. Simplified higher education teaching production process outlined in 2.2.1, synthesised from the literature. Based on (Hanushek, 1979; Johnes, 2006; Jongbloed et al., 2018; Marginson, 2008; Sampson, 2000; Sampson & Froehle, 2006).

In summary, HEIs use institutional resources alongside heterogeneous and partly unobserved student inputs to produce generally latent human capital. Because this output cannot be directly observed it is measured through proxies such as degrees or research output. Due to selection effects, the temporal misalignment of inputs and outputs, and the joint production of teaching and research, simplified modelling choices and careful sample construction are required. As many variables cannot be observed and data availability is limited, the model must remain very simple. The purpose is not to estimate the full educational production function but a feasible and representative approximation of teaching output.

2.2.2 Quality measurement in higher education

Section 2.1 considered the differences in how quality is evaluated and constructed in different types of services and this section will go over quality measurement and evaluation in higher education. While the focus of this thesis is on efficiency, Grönroos & Ojasalo (2004) argue that productivity improvements in services cannot be evaluated independent of quality as, unlike manufacturing, in service production changes to production input or delivery mode directly affect customer experience and outcomes. Therefore, quality and efficiency are interdependent and efforts to change one may have effects on the other. Quality in services is also more intangible and harder to quantify especially when the relative importance of functional quality increases (Grönroos, 1984; Parasuraman et al., 1988). Because quality is partly latent, partly experiential, and partly reputational, it is difficult to incorporate directly into comparable productivity models. The purpose of this section is not to solve quality measurement, but to define the limits within which throughput-based efficiency estimates can be interpreted.

To explain why technical quality is particularly difficult to evaluate in higher education, it is useful to consider another way of classifying products. Goods and services can be classified into search, experience, and credence goods originating in Nelson's (1970) distinction between search and experience goods, later extended by Darby & Karni (1973) through the concept of credence goods. Search goods can be evaluated before purchase, experience goods during or after consumption but credence goods are difficult or impossible to evaluate even *ex post*. Higher education exhibits credence-good characteristics, although some aspects of the student experience are observable during consumption (Darby & Karni, 1973; Jongbloed et al., 2018; Nelson, 1970). Students may assess, for example, whether teaching is engaging, administrative services work smoothly, and if campus life is satisfactory (Jongbloed et al., 2018). It is much less clear how these factors translate into learning gains relative to counterfactual or how much their labour market success is attributable to the institution as outcomes emerge gradually through

career and life-course processes (Angrist et al., 2017; De Witte & López-Torres, 2017; Johnes et al., 2017; Jongbloed et al., 2018).

Figure 1 showed that as customers are more engaged in the production process, their increased exposure leads them to value functional quality. At the same time, the credence nature of higher education makes technical quality difficult to evaluate. Students can observe many aspects of the educational production process during schooling, but the value of specific knowledge or skills is difficult to appraise and realised over time. This creates a distinction between the observable experience of education and the more difficult question of marginal student development produced.

The conceptual ideal of technical quality of teaching would be to observe the institution's causal contribution to student development (Angrist et al., 2017; Hanushek & Ettema, 2017). In practice this is extremely difficult to isolate and approximated through outcomes measures such as placement in labour market or academia (Jongbloed et al., 2018; Lee & Johnes, 2022). Such a measure leads to attribution problems between student and institution and changes in labour market outcomes often reflect confounders such as market dynamics rather than changes in quality (Angrist et al., 2017; De Witte & López-Torres, 2017; Hanushek & Ettema, 2017). Because higher education has credence-good characteristics (Jongbloed et al., 2018), early labour market outcomes are partly shaped by reputation as they also reflect social beliefs about quality, not just quality itself (Espeland & Sauder, 2007; Spence, 1978). Reputational effects in turn are partly shaped by institutional production conditions, including peer effects and networking opportunities, but they also affect degree value in ways that are not fully explainable by student learning outcomes (Angrist et al., 2017; Jongbloed et al., 2018; Spence, 1978). Consequently, adjusting educational outputs for quality is challenging because institutions' causal contribution is difficult to isolate, and observed outcomes beyond degrees reflect selection, peer, and institutional reputation effects in addition to educational efficacy.

Measurement problems are intensified when the service output is transformation of the customer and in these situations functional quality is often the main observable dimension (Grönroos, 1984). Due to the large customer role in education and the transformational service delivery mode, functional quality may lead to technical quality by improving important customer-provided inputs to production through increased motivation and engagement (Bruinsma & Jansen, 2007; Sampson & Froehle, 2006). This shows the value of functional quality in service production and why producers forego potential efficiency gains to preserve it (Frei, 2006). In services with significant customer participation, such accommodations may be a part of production rather than deviations from it (Frei, 2006; Grönroos & Ojasalo, 2004).

High salience of functional quality does not necessarily imply decreased customer interest in technical quality (Grönroos, 1984). The technical service quality in a transformational, high-participation service such as education may even be more important. As customer participation increases, they use more resources towards the outcome which has large long-term consequences. Therefore, the quality of the outcome is of central importance even when students cannot evaluate it precisely during or immediately after production.

Due to the difficulty in approximating technical quality and the limitations of cross-institutional quality comparison, internal monitoring of functional quality is one practical tool for quality evaluation. Universities use student surveys, tracking multiple aspects of functional quality (Jongbloed et al., 2018). While these may have some use, student evaluations of teaching are often weakly related or even opposed to technical quality depending on student expectations (Uttl et al., 2017). Meta-analyses show that student evaluations of teaching correlate weakly or are even uncorrelated with objective learning outcomes while systematically influenced by grading leniency, student expectations and course difficulty (Marsh, 2007; Uttl et al., 2017).

Because degree value is multidimensional, students may differ in how they weight components between signalling and human capital improvement creating diverging expectations and preferences (Spence, 1978). When student expectations diverge from institutional goals in education, functional quality metrics may encounter a Goodhart-type problem where their optimisation rewards choices that improve short-term satisfaction without improving or even harming the underlying educational objective (Goodhart, 1984). This is also a risk of education efficiency measurement more broadly if a specific, limited model is treated as a metric to maximise (Vakkuri & Meklin, 1998).

The characteristics of educational quality connect to a broader paradox in service operations. Efficiency can be improved through standardisation, automation, and associated reduction of variation and customer contact, yet in many services, accommodating customer-induced variation is itself central to value creation (Chase, 1978; Frei, 2006; Lillrank & Liukko, 2004; Silvestro et al., 1992). Applied to higher education, more standardised delivery may improve consistency, scalability, and resource efficiency, but some dimensions of educational quality may depend on responsiveness to heterogeneous student needs, richer interaction, or flexible pedagogical approaches. Generally teaching delivery is more standardised in early stages of higher education with the routineness declining as students develop more skills and require more specialised instruction to continue their development. The relationship between efficiency and quality is therefore neither necessarily antagonistic nor harmonious. Process redesign can potentially improve both, for example

by removing administrative waste or standardising low-value variation but when compressed activities are central to learning, quality and efficiency may be in real tension.

In summary quality in education is difficult to measure comprehensively and cannot be reduced to a single observable indicator. Quality can be incorporated into productivity measurement through indicators such as student satisfaction, labour-market outcomes, or other quality-adjusted outputs. However, these do not provide a satisfactory approximation of underlying educational quality for cross-institutional benchmarking. Functional quality measures capture only the observable experience of education and are influenced by student expectations and institutional context, while technical quality is difficult to isolate from selection, peer, reputational, and labour-market effects. Due to the limited value of available quality indicators, outputs are not adjusted for quality, and this thesis focuses exclusively on pure throughput analysis.

2.3 Efficiency and productivity in higher education

This section connects the preceding service and higher education arguments to efficiency and productivity measurement. It first discusses productivity measures and explains their limitations in higher education benchmarking. It then introduces frontier efficiency methods as a response to the aggregation and weighting problems posed by productivity measurement in higher education. Subsequently, this section reviews previous research in the field to inform methodological choices. Finally, this section reviews empirical findings from higher education efficiency studies to later compare results and trends from previous research to the findings of this thesis.

2.3.1 Limitations of productivity measurement methods in higher education

Productivity measurement is a common approach to assess operational performance. Fundamentally, productivity represents the relationship between inputs and outputs, generally expressed as a ratio of output to input (Fried et al., 2008). Productivity growth allows more outputs to be produced with the same inputs or equivalently the same outputs produced with fewer inputs (Syverson, 2011). The previous section showed the difficulty in defining and measuring the inputs and outputs of higher education teaching. For example, as we cannot reliably observe student learning, productivity measurement relies on proxies such as degrees which can be measured against resource use. As HEIs have multiple inputs and outputs, the selection and weighting of these variables is central to efficiency or productivity measurement.

Single-factor productivity is a form of partial productivity and represents the relationship between output and a single input, such as labour (Kendrick, 1956; Syverson, 2011). Index numbers extend the same ratio-based logic by comparing relative changes in inputs and outputs, thereby reducing a complex production process into simple scalar measures (Diewert, 1992). Common forms of this in education are teaching personnel per student or graduates per unit of spending (Hanushek & Ettema, 2017). These metrics are simple and can often be constructed from limited data, but they still require decisions about which inputs and outputs to include and how to weight them. For example, it is not clear what teacher to student ratio is optimal or whether this is variable between institutions (Goldhaber & Brewer, 1997; Hanushek & Ettema, 2017). The aggregation into individual scalars also masks sources of differences in outcomes (Syverson, 2011; Vakkuri & Meklin, 1998).

The simplicity of partial and index-based productivity measures makes them suitable for measurement in large data sets or between heterogeneous units, though this characteristic can lead to bias due to omitted inputs and substitution (Abramovitz, 1956; Solow, 1957). For example, comparing HEIs on students per staff without considering expenditure, field composition, and research obligations would be misleading. This motivates multi-factor productivity approaches (often termed total factor productivity or TFP) which relate output to a bundle of inputs. This is theoretically attractive as they reduce the omitted-input problem of partial productivity measures, but they require assumptions about input and output measurement, aggregation and weighting. These assumptions are difficult to satisfy particularly when outputs are not traded at market prices or quality differences are central to the service outcome (Grönroos & Ojasalo, 2004; Hanushek & Ettema, 2017; Vakkuri & Meklin, 1998). Once accounting for multiple inputs and outputs, the problem shifts to how they can be aggregated or weighted in a defensible way.

Previously outlined characteristics of service production generally and higher education specifically: heterogeneous customer inputs, multiple intangible outputs, and non-standard processes limit the applicability of traditional productivity measures based on aggregate ratios or value-based accounting (Grönroos & Ojasalo, 2004; Hanushek, 1979; Sampson & Froehle, 2006). Table 2 summarises how many assumptions in standard productivity measurement are difficult to satisfy in teaching productivity measurement. The next section will show why efficiency measurement is a logical response to these challenges and the appropriate focus for this thesis.

Table 2. Conventional productivity measurement assumptions and corresponding challenges in measuring teaching productivity.

Measurement issue	Standard productivity measurement requirement	Challenge in teaching productivity
Output measurability	Outputs can be counted or aggregated (Diewert, 1992; Hulten et al., 2001; Triplett & Bosworth, 2004).	Outputs are individualised, partly intangible, and delayed (Hanushek & Ettema, 2017; Massy et al., 2013).
Output valuation	Prices approximate relative value (Diewert, 1992; Hulten et al., 2001).	Value develops over time, and outputs cannot be fully defined ex ante. Often sold in bundles rather than itemized (Grönroos & Ojasalo, 2004; Sampson & Froehle, 2006).
Input attribution	Inputs are separable and measurable (Abramovitz, 1956; Solow, 1957).	Staff, facilities, admin, research etc. are shared across activities (Massy et al., 2013).
Process comparability	Similar inputs generate comparable outputs through stable technology (Charnes et al., 1978; Fried et al., 2008; Solow, 1957).	Teaching processes vary by programme, cohort, and participation (Frei, 2006; Sampson & Froehle, 2006).
Input-output boundary	Inputs are provided by the producer and outputs by the production process (Sampson & Froehle, 2006; Triplett & Bosworth, 2004).	Value is created through interaction in a non-routine fashion (Grönroos & Ojasalo, 2004; Lillrank & Liukko, 2004; Sampson & Froehle, 2006).

2.3.2 Efficiency measurement and methods

The limitations presented in Table 2 do not make input-output comparison irrelevant, but they indicate that fixed productivity ratios are insufficient standalone measures for benchmarking institutional performance. Instead of showing how much output is produced per unit of input, efficiency measures how well an institution transforms inputs into outputs compared to what peers have achieved. Farrell (1957) formalised this idea through a frontier approach to efficiency measurement. Instead of comparing institutions against predefined ratios or externally imposed weights, frontier methods construct a best practice frontier from observed performers. Efficiency is therefore an inherently relative concept as institutions are benchmarked against each other rather than a universal productivity ratio.

While productivity refers to observed input-output ratios, efficiency is defined by radial distance from the technologically feasible frontier derived from observed best practices among benchmarked firms (Farrell, 1957; Fried et al., 2008). Therefore, efficiency scores of frontier methods represent relative conditional efficiency among a selected sample rather than a universal ratio (Dyson et al., 2001).

Building on Farrell's frontier approach, Charnes et al. (1978) introduced data envelopment analysis (DEA) which constructs a piecewise linear production frontier through linear programming. DEA accommodates multiple inputs and outputs, does not need price information and allows efficiency to be measured without approximating a functional form or aggregating outputs to a single scalar (Boussofiane et al., 1991; Charnes et al., 1978; Worthington, 2001). Another common frontier method is stochastic frontier analysis (SFA) introduced by Meeusen & van den Broeck (1977) and Aigner et al. (1977).

Several other alternatives are available for higher education productivity analysis, but they are not as suitable as frontier methods. The previously mentioned ratio and index-based approaches are transparent and useful for diagnostics but insufficient. Multi-factor productivity incorporates multiple variables and tracks productivity change but requires aggregation and weighting assumptions that are difficult to justify when outputs are not priced. Regression-based production models can estimate average input-output relationships, but they require a specified functional form and are less directly suited to identifying best-practice benchmarks.

For frontier methods the choice is between parametric and non-parametric with SFA and DEA respectively. As a parametric method requiring a predefined production function, SFA includes an error term in addition to inefficiency allowing the distinction between technical inefficiency and

measurement error or random shocks (Aigner et al., 1977; Meeusen & van den Broeck, 1977). The downside of SFA in education is that production technologies across institutions are heterogeneous and difficult to estimate while the assumptions on functional form and the distributions of inefficiency and errors are significant and rarely testable (Hanushek, 1979; G. Johnes & Johnes, 2009; Worthington, 2001). By comparison, standard non-parametric DEA does not require a defined production function and thus attributes all variation to frontier efficiency making it sensitive to measurement errors and outliers (Simar & Wilson, 2000). The lack of a stochastic error term means that noise is not separately modelled, though this also avoids making assumptions on the distribution of errors (Fried et al., 2008; Simar & Wilson, 2000). To compensate, statistical inference necessitates the use of additional methods such as bootstrapping, to obtain bias-corrected estimates and assess statistical precision (Simar & Wilson, 2000, 2007). Given the exploratory nature of this thesis and the difficulty in defining a justifiable and modellable production function in higher education, DEA is the method of choice for this research.

Having established DEA as the preferred framework for cross-sectional analysis, frontier methods can also be used to analyse efficiency change over time. The Malmquist productivity index (MPI) extends DEA by measuring changes in total factor productivity between periods through comparisons of successive production frontiers (Färe et al., 1994; Fried et al., 2008). It decomposes productivity change into efficiency change and technological change. Efficiency change represents institutional movement relative to the frontier, while technological change is shifts in the frontier itself (Färe et al., 1994). Consequently, MPI complements cross-sectional DEA by distinguishing whether productivity improvements arise from institutions catching up with existing best practice or from expansion of the production frontier.

In summary, DEA is well suited for higher education as it accommodates the production characteristics better than many conventional measures. Education is multi-input, multi-output (Johnes, 2006), its outcomes are shaped by heterogeneous customer inputs and environments (Hanushek, 1979; Sampson & Froehle, 2006), and its exact production function is difficult to specify. Limited data availability makes strict parametric modelling challenging and limits inference to a rough estimate (De Witte & López-Torres, 2017; Hanushek, 1979; Massy et al., 2013). This is where DEA is useful as it limits assumptions and does not require a functional form (Charnes et al., 1978; Dyson et al., 2001). Simultaneously it is important to recognise the model does not translate imperfect data into perfect results. DEA efficiency results are conditional on the selected variables, sample, and modelling assumptions and do not represent true productivity, and should therefore be interpreted as structured estimates of relative performance (De Witte & López-Torres,

2017; Dyson et al., 2001; Johnes, 2006). MPI and productivity ratios provide detailed analysis of productivity development over time.

2.3.3 DEA applications in higher education

Efficiency research in higher education usually focuses on teaching efficiency. This reflects its centrality to the mission, importance to key stakeholders, and relative measurability. While imperfect, inputs and outputs such as staff, enrolments and degrees map more cleanly to educational activities compared to research indicators (Hicks et al., 2015; Moed, 2005). As a result, most frontier-based analyses on higher education focus on teaching efficiency (Johnes, 2006; Worthington, 2001). Quantitative benchmarking studies generally use DEA or SFA (De Witte & López-Torres, 2017; Worthington, 2001).

Johnes (2006) documents that the applications of DEA on higher education were motivated by concerns over teaching efficiency, capital allocation and institutional accountability. Governments and funding agencies have incentives to monitor the output of public institutions, and those institutions are motivated to appear efficient to justify continued public funding (Jongbloed et al., 2018; Vakkuri & Meklin, 1998). Such funding is often closely tied to teaching related variables like student throughput (Geuna & Martin, 2003; Pruvot & Estermann, 2022). The omission of quality and long-term knowledge sharing dimensions is acknowledged but this framing provides a feasible framework for benchmarking institutions operating under similar conditions (De Witte & López-Torres, 2017; Doherty, 2008).

Exact input-output formulations vary considerably between studies, although there are some core variables which are included most of the time (De Witte & López-Torres, 2017). For inputs, these include some combination of number of academic staff, full-time staff numbers and instructional expenditure or total operating budgets. Students are sometimes treated as an input, sometimes an output and in other cases as intermediate products, reflecting different interpretations of the production process (Johnes, 2006; Kao, 2009). Treating students as an input can emphasise selectivity and quality while holding them as an output may reward throughput and completions. Using operating instead of personnel budgets may obscure underlying differences in pedagogical efficiency by including for example support staff and varying space-related costs, but it also ensures that all resource use is included in calculations (De Witte & López-Torres, 2017). Some researchers prefer to use staff headcount instead of spending as adjusting for local salary differences is difficult (Andersson & Sund, 2022).

For teaching outputs, degrees awarded is the most common due to their broad availability and relevance (De Witte & López-Torres, 2017). Alternative measures include course completions and credits accumulated. While these measures theoretically have more relevance and better time alignment to inputs, they suffer from poor availability in cross-institutional comparison (Räty & Kivistö, 2006). Some research uses placement in industry or academia as output (Lee & Johnes, 2022), though this suffers from attribution problems as previously discussed.

Ignoring research activity entirely when benchmarking teaching risks attributing larger research output or focus to teaching inefficiency (Johnes & Johnes, 2009; Johnes & Yu, 2008). Some studies attempt to accommodate for this by modelling both teaching and research efficiency jointly, for example (Abbott & Doucouliagos, 2003; Andersson & Sund, 2022). This however increases model complexity and data requirements while increasing sensitivity to variable selection. It may also make inference more difficult as the production of research outputs and doctoral degrees differs temporally from bachelor's and master's degrees. Other researchers model teaching and research separately, for example (Herberholz & Wigger, 2021), though this still leaves the relative weighting of these activities unresolved.

New literature is also trying to incorporate quality metrics into DEA models to increase the precision of teaching efficiency estimates. Measures such as student satisfaction, completion rates and labour market outcomes have been applied as proxies for quality in addition to throughput metrics (Lee & Johnes, 2022). This can materially change efficiency rankings, highlighting potential trade-offs between quantity and quality (Lee & Johnes, 2022). However, as previously discussed, such metrics introduce potential bias, endogeneity and measurement error particularly when they reflect student characteristics or prestige rather than institutional quality (Angrist et al., 2017; Doherty, 2008; Goldhaber & Brewer, 1997; Utzl et al., 2017).

Due to data constraints, benchmarking is usually performed at the institution level, aggregating all schools and departments of a university to a single entity (De Witte & López-Torres, 2017; Herberholz & Wigger, 2021; Johnes et al., 2017; Worthington, 2001). The downside is that this leads to significant heterogeneity due to differences in institutional composition (Agasisti & Bonomi, 2014; Bruni et al., 2020; Herberholz & Wigger, 2021). Some fields have higher resource demands than others, with engineering and medical fields for example often requiring significantly more infrastructure compared to social sciences (Herberholz & Wigger, 2021). Practically this bias can be alleviated by limiting the sample to schools of a single field (Kong & Fu, 2012), or through clustering methods where institutions are benchmarked within similar groups (Herberholz & Wigger, 2021). Agasisti & Bonomi

(2014) use a particularly involved approach where they decompose universities to their individual schools, deriving efficiency scores for those schools and aggregating individual scores into a composite score for each institution.

Efficiency research has often focused on national benchmarking (Abbott & Doucouliagos, 2003, 2009; Johnes, 2006, 2008; Johnes & Yu, 2008), though cross-country comparisons have increased more recently (Agasisti & Johnes, 2009; Agasisti & Pérez-Esparrells, 2010; Andersson & Sund, 2022; Herberholz & Wigger, 2021; Parteka & Wolszczak-Derlacz, 2013). One potential explanation may be that national studies can more readily combine registry and institutional data under consistent reporting frameworks, enabling richer and more comparable variable selection across institutions. Cross-national benchmarking has comparatively poor data availability and additional heterogeneity of units due to environmental differences. HEIs' positions in national regulatory and funding regimes shape and individualise feasible input-output combinations and institutional incentives (Agasisti & Johnes, 2015; Marginson, 2008).

2.3.4 Trends in higher education productivity research results

Empirical applications in higher education often find modest average productivity change, but the decomposition varies across countries, periods and model specifications. For example, Johnes (2008) reports productivity growth in English higher education driven by technological change but offset by declining technical efficiency, while Parteka & Wolszczak-Derlacz (2013) use bootstrapped Malmquist indices to show that many estimated changes are statistically uncertain. There are often significant differences in efficiency across institutions (Johnes, 2006; Worthington, 2001).

In the European context these relative efficiencies and institutional standings also tend to remain stable over time, especially for institutions near the efficient frontier (Johnes, 2008; Parteka & Wolszczak-Derlacz, 2013). The results may reflect the slow-moving nature of universities, and persistent institutional characteristics such as mission and funding, but also suggest unrealised potential for some institutions. At the same time stable rankings may partly reflect the methodological properties of frontier measurement and limited data availability as estimated efficiency is conditional on selected variables, peer groups, and frontier specification (Fried et al., 2008; Simar & Wilson, 2007).

Technological change, measured by frontier movement which represents the efficiency of the most efficient institutions is often modest (Johnes, 2008; Parteka & Wolszczak-Derlacz, 2013). This may reflect the institutionally constrained and labour-intensive nature of higher education and professional

services broadly (Sampson, 2018; Susskind & Susskind, 2017). Still, change over time is mostly driven by frontier movement rather than technical efficiency change, meaning institutions mostly move together in accordance with system shifts rather than through large differences in managerial catch-up. Institutional rankings, particularly at the frontier do not change much over time. This does not necessarily imply an absence of innovation as many changes may affect mainly quality or student experience which are usually left unmeasured. Other studies also find more growth over time though it is often regional (Agasisti & Johnes, 2009), or requires more comprehensive measurement such as the accounting of research contributions (Worthington & Lee, 2008).

Significant change may often originate from exogenous factors such as shocks, policy changes or student cohort effects in addition to organisational innovation or restructuring (Agasisti & Johnes, 2015; Fried et al., 2008; Marginson, 2008). Managerial options can be limited due to low technological growth, and especially in public systems constrained further by institutional mandates and incentive structures which limit the scope and payoff of change.

The findings of modest productivity growth and limited frontier movement draw the invocation of Baumol's (1967) "cost-disease". Baumol suggested that as services are labour-intensive and resistant to automation, they have limited avenues for productivity improvement. That is, they are not as readily able to take advantage of technological progress due to their inherent reliance on human labour. Baumol's view has since been criticised for being overly pessimistic. While technological advances have driven significant productivity growth in manufacturing, the service sector has had periods where its productivity growth has outperformed goods-production (Triplett & Bosworth, 2004).

However, the productivity growth within services has been asymmetric and often dependant on service characteristics as some are more able to substitute capital for labour than others. As discussed previously, services with particularly high customer participation have been more constrained in efficiency potential as personalised customer interaction is central to their production process (Frei, 2006; Lillrank & Liukko, 2004; Sampson & Chase, 2022; Sampson & Froehle, 2006). Some services such as education and healthcare also exhibit low price elasticity of demand and are heavily regulated, which further limits their incentives and opportunities for productivity growth (Triplett & Bosworth, 2004). Consequently, high customer participation, personalised services in particular, have seemingly encountered Baumol's cost disease.

The cost disease has hit some services harder than others. Professional services such as healthcare, consulting and higher education have experienced relatively little technological productivity gains (Sampson, 2018; Susskind & Susskind, 2017). Due to their reliance on educated, well-paid employees, professional services could theoretically obtain large benefits from labour-saving technology, but this has proven difficult to implement in practice, often due to task complexity or integration (Sampson, 2018; Sampson et al., 2023). As these services derive much of their value through customer interaction, the employee time is a large component of the product (Lewis & Brown, 2012; Sampson et al., 2023; Sampson & Froehle, 2006).

A common way to organise professional services is to standardise and out-source routine tasks to lower-skilled junior workers while limiting expensive specialist work to tasks that require their expertise (Lewis & Brown, 2012; Lillrank & Liukko, 2004; Susskind & Susskind, 2017). Sampson et al. (2023) study ways to use technology to expand the number of these tasks to reduce the workload of professionals and save costs. However, as they point out, practical application often depends on interactions between tasks, for example compiling teaching material and presenting the material in class and leading discussion. In practice universities use task offloading by utilising teaching assistants to respond to student queries, grade some assignments or lead informal practice sessions but there are also often institutional minimums for professor involvement in teaching limiting potential of different arrangements to uphold standards.

In the presence of limited options, a potential path for universities to increase efficiency is through scale. Teaching efficiency often shows increasing returns to scale, especially for small institutions (G. Johnes & Johnes, 2009; Salas-Velasco, 2024). While larger cohorts may allow for more efficient use of fixed resources, they can also increase the variability between student inputs and general administrative burden (Agasisti & Johnes, 2015; Humlum & Thorsager, 2021; G. Johnes & Johnes, 2009; Lazear, 2001; Winston, 1999; Winston & Zimmerman, 2004). Consequently, positive scale returns in education eventually plateau or reverse (Agasisti & Johnes, 2015; G. Johnes & Johnes, 2009), making scale a limited tool for improvement, though its potential may be increased through technology. Vaarma & Li (2024) use machine learning to predict student drop out risk in Finnish higher education, finding that such approaches aid in identifying students in need of additional support, which may provide efficiency gains in student services.

As customer heterogeneity is a significant factor for efficiency potential in services with high customer participation, scale increases may cause declines in efficiency or quality as gaps in student ability and expectations widen. Reducing the variability in student inputs, for example by sorting students on

their ability level or motivation could improve mapping between teaching and student expectations. Ability grouping in education has some positive evidence at lower levels (Steenbergen-Hu et al., 2016), and peer effects are significant in higher education (Humlum & Thorsager, 2021; Lazear, 2001), but institutional control may be limited in their formation (Carrell et al., 2013). Higher education admissions requirements, different learning tracks which may have cutoffs, and required or recommended prerequisites for courses theoretically reduce student heterogeneity and improve peer group formation, but their causal impacts on outcomes are uncertain. Balancing heterogeneity, quality, and access is a difficult challenge in higher education. Institutions must be careful as optimal scale is conditional on institutional environment and mission and often constrained by multiple factors, where scale increases may cause subsequent bottlenecks in throughput or degrade quality.

2.4 Conceptual framework on the higher education teaching process measurement

Services can be defined by their reliance on customer-provided inputs (Sampson & Froehle, 2006). Some services resemble manufacturing in their standardisation (Chase, 1978), as their production can mostly be executed without customer involvement and customer participation is relegated to the delivery stage, manifesting as selection from tailored options. Other services are more reliant on customer-provided inputs as those are necessary components of production. Some services including higher education not only depend on customer inputs but also solve such individual and complex problems that their production process is co-defined by the customer.

The level of customer participation is a key determinant of efficiency potential as the earlier customers become involved in the producer's value chain (Figure 3), the less the producer can plan and execute production without the customer's involvement. This limits the standardisation potential of the production process and causes value to be found increasingly in individualisation and interaction rather than embedded in the final product. This means that efficiency is not strictly dependent on the ability to produce output but is more dependent on process and extended customer interaction in ways that often elude measurement.

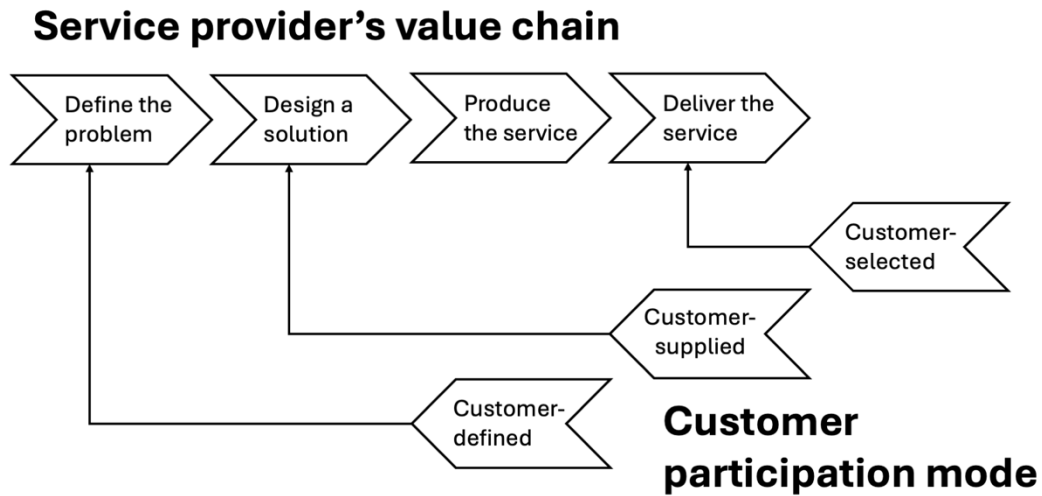


Figure 3. The intersection of the customer in the provider's value chain depending on the service's primary mode of customer participation.

Productivity measurement begins by defining relevant inputs and outputs which is difficult when the customer is also a key input provider and an important agent through the production process (Sampson, 2000; Sampson & Froehle, 2006). Consequently, higher education production inputs and processes are non-standard and conditional on their student body, institutional characteristics, and environment (Hanushek, 1979). Outputs vary in quality in subtle ways and their contribution is difficult to separate between institution and student (Belfield & Fielding, 2001; Hanushek & Ettema, 2017). These challenges motivate the use of frontier methods for relative efficiency in a selected sample rather than absolute productivity ratios because they accommodate multiple inputs and outputs without requiring explicit aggregation (De Witte & López-Torres, 2017; Fried et al., 2008). While these methods do not solve the entire measurement problem, they allow comparison of multiple inputs and outputs. The main remaining challenges and limitations caused by higher education characteristics are outlined in Table 3.

Table 3. Service theory implications for higher education productivity measurement.

Service theory construct	HEI implication	DEA implication
Customer provided inputs (Sampson, 2000; Sampson & Froehle, 2006)	Student ability, effort, prior preparation, motivation shape production	Inputs are partly unobserved
Customer participation during production (Chase, 1978; Sampson & Froehle, 2006)	Teaching outcomes depend on interaction, progression, supervision, institutional support	Production process is heterogeneous
Intangible/credence output quality (Grönroos, 1984; Zeithaml, 1981)	Quality is not fully captured by enrolments/graduates	Outputs are throughput proxies
Environmental heterogeneity (Agasisti & Johnes, 2015; Parteka & Wolszczak-Derlacz, 2013)	Funding, regulation, tuition, study rights, selection, and field mix differ across countries and institutions	Efficiency is conditional on sample and environment

Due to its limitations, higher education benchmarking has been criticised for giving decision makers unfounded confidence and misplaced focus (Elton, 2004; Goodhart, 1984; Morrison et al., 1995). While methodological extensions have increased the robustness of results (Dyson et al., 2001; O'Donnell et al., 2008; Simar & Wilson, 2000), researchers acknowledge there is a fine line between useful diagnostic tool and misdirection which gives decision-makers a narrow lens for performance evaluation (Fried et al., 2008; Johnes, 2006; Worthington, 2001).

The central implication is that frontier efficiency estimates must be treated as conditional and relative. They measure the ability of institutions to transform observable inputs into observable outputs given their characteristics and environment, not absolute managerial effectiveness or value production. However, as governments evaluate public universities largely on their ability to produce output, typically measured in graduates, throughput analysis is a necessary tool and an important indicator for leadership.

The following analysis should be interpreted as a study of observable teaching throughput rather than a comprehensive measure of educational value creation. While different specifications were trialled, these represent only a comparatively narrow subset of possibilities to measure higher education teaching efficiency.

3 Methodology

This thesis uses a quantitative frontier-benchmarking design to evaluate the teaching efficiency of Aalto BIZ relative to selected European higher education institutions. The design consists of a stationary DEA model which estimates both the current relative teaching efficiency of these HEIs and a longitudinal productivity analysis to assess productivity trends over time.

3.1 Research design

The underlying managerial objective necessitating this thesis is to develop understanding of Aalto BIZ's teaching efficiency compared to its peers. University management desires to have perspective on resource use and its returns at Aalto BIZ. Additionally, the development of teaching productivity at Aalto BIZ is of institutional interest. Informed by previous literature and the limitations of higher education measurement, the objective of this thesis is to develop a feasible benchmarking approach for assessing teaching efficiency at Aalto BIZ to generate managerial insights into Aalto BIZ's relative performance and recent productivity development.

A key managerial objective of productivity research is to find the optimal mix of inputs which produces the maximum number of valuable outputs. Such problems call for optimisation methods (Kopcsó & Pachamano, 2018), but the previously outlined characteristics of higher education limit their usefulness. The heterogeneity and incomplete observability of the production function support the use of relative benchmarking with frontier methods (De Witte & López-Torres, 2017; Hanushek, 1979; Johnes, 2006). These methods can estimate the benchmark either as the most efficient performers in the data given observed production technologies (DEA) or theoretical best performance with a specified production function (SFA) (Aigner et al., 1977; Charnes et al., 1978; Fried et al., 2008).

Alternative approaches worth considering for the research problem are simple performance indicators or partial productivity such as spending per graduate (García-Aracil & Palomares-Montero, 2010). However, without an optimisation framework, such as DEA, that endogenously determines the relative importance of multiple inputs and outputs, any overall comparison rests on subjective weighting decisions. Another possibility is to use regression to calculate what is associated with spending or graduation rates, but these measure average relationships rather than best practices (Worthington, 2001). Regression can incorporate observed heterogeneity, but its interpretation remains limited when important institutional and student characteristics are unobserved. Restricting the sample to more comparable institutions may

reduce heterogeneity but also reduces the number of observations available for estimation (Hanushek, 1979).

Frontier methods by contrast evaluate performance relative to best observed or estimated practice. DEA, originally developed by Charnes et al. (1978) is a non-parametric linear optimisation method for estimating the relative efficiency of decision-making units (DMUs). Unlike regression, which estimates average relationships, DEA constructs an empirical frontier of the most efficient units and evaluates other units by their distance to this frontier. The efficiency scores are obtained from a weighted ratio of a unit's outputs to inputs, where the weights are optimised with other

The empirical analysis will consist of two complementary parts. First, a stationary DEA comparison of relative efficiencies across all sampled institutions including Aalto BIZ position and potential field differences. Second, a year-by-year benchmark over time to evaluate productivity development in higher education and how it compares to the development of Aalto BIZ. The empirical analysis combines Python and R. Python was used for data preprocessing while R was used to estimate DEA models, bootstrap bias corrections, and Malmquist productivity indices. The formal DEA model specification is presented in section 3.3.

3.2 Data collection and handling

The data are primarily sourced from European tertiary education register (ETER). Their publicly available database provides harmonised institution-level panel data on European HEIs (ETER, 2025). As ETER aggregates data to the university rather than school level, comparative data on Aalto BIZ was sourced from internal school resources. An initial sample of business schools was collected based on data availability and their similarity to Aalto BIZ. This included business schools some of which also had business law and appeared in Financial Times rankings or were accredited. The aggregation at the university level proved to be a significant limitation as business schools, including Aalto BIZ, are generally a part of large universities with varying field mixes which makes comparison difficult. The data set was also incomplete, and some institutions had to be dropped as they were missing essential data. Among large countries, French, Italian, and Spanish universities have a lot of missing data on ETER and thus could not be included. Due to these limitations, the final business school sample consists of 19 HEIs, largely from Germany, the Nordics and Eastern Europe. These institutions are small compared to universities with multiple schools, and they include both private and public schools.

While the sample of business schools is adequate for a simple DEA model, a larger sample improves frontier coverage and discrimination, provided the added DMUs remain comparable (Dyson et al., 2001). Simultaneously, this thesis has previously established that universities are heterogeneous with heterogeneity increasing between mission. As such it is important to balance the gains from additional sample size with potential bias from increasing heterogeneity. As a result, a clustering approach is employed to preserve discrimination between potentially differing institutions. An additional 185 institutions were selected from ETER utilising a similar protocol as the selection of business schools. Institutions were identified through manual screening of the ETER database. Selection prioritised institutions with business and social science orientations because previous literature suggests subject-oriented peer groups are more comparable than comparisons with comprehensive universities (Agasisti & Bonomi, 2014). Institutions with substantial missing data for the selected variables were excluded. The final sample was then grouped into four clusters according to their disciplinary profile. In the end, cluster 1 consists of the 19 business schools, cluster 2 includes 40 HEIs which focus on business and social science, cluster 3 has 73 institutions with a broader social science focus, and cluster 4 includes 78 generalist institutions which may include engineering and medicine. In total there are 214 HEIs in four clusters.

ETER data are available for the period 2011-2022. To reflect the multi-year nature of higher education production and reduce the influence of year-to-year fluctuations, stationary efficiency comparisons use five-year averages. The reference period is defined by the input years. Input variables and intermediate student measures are averaged over 2015-2019, representing the most recent five-year input window that permits evaluation under all considered lag specifications (0-3 years).

Degree outputs are lagged because they result from resources and enrolled student stocks observed in earlier years. Lags from 0-3 years were estimated as robustness checks but the exact value did not materially affect results. A two-year lag was selected as the preferred specification because it provides a reasonable balance between representing the time required for educational production and retaining the most recent available observations. It broadly corresponds to the target duration of a master's programme, although it does not map to individual cohorts exactly and is slightly shorter than the nominal duration for most bachelor's programs. Doctoral programmes take considerably longer, but doctoral education accounts for a relatively small share of university teaching activity.

Under the selected two-year specification, the stationary analysis combines resource inputs and enrolled students from 2015-2019 with degree outputs

from 2017-2021. A small number of DMUs have missing observations which are partially backfilled using data from earlier years for stationary comparison. This contains the assumption that production technology is comparable over time and slight temporal differences do not bias results.

Later year-by-year and Malmquist productivity analyses apply the same two-year lag selected for stationary comparisons. Consequently, the input data for these is between 2011-2020 corresponding to degree outputs from 2013-2022. This allows the dynamic analyses to report the results from the most recent possible period while maintaining a consistent lag structure.

3.3 Model specification

The empirical analysis was conducted in R using the deaR package. The way DEA assigns weights to variables means that increasing the number of these variables in the model also increases potential combinations of weights leading to full efficiency. This is exacerbated by small sample relative to number of variables. Doing so results in the frontier becoming disproportionately populated and the model losing discrimination between units as there are too many possible efficient combinations relative to the total number of DMUs. Any unit for which its ratios of inputs to outputs are most efficient in the set will appear efficient and assuming that these are evenly distributed across the sample, there will be a number of efficient units equal to the product of the numbers of inputs and outputs. Therefore, a rule of thumb is to have $2 \cdot m \cdot n$ DMUs where m is the number of inputs and n is the number of outputs. Pruning a DEA model is therefore desirable as it increases discrimination and decreases the chance of unimportant variables influencing efficiency scores while including a set of variables that adequately captures the production process (Dyson et al., 2001). This is particularly relevant when considering cluster-level analysis where the sample is reduced.

There is broad agreement on using degrees as outputs of higher education (De Witte & López-Torres, 2017). Therefore, the model includes graduates at bachelor's and master's levels. One of the most important and overlooked input variables in education productivity are the students themselves. They are usually treated as an output, but this thesis treats them as an intermediary variable in a two-stage model where they are both an output of resource use and an input into the production of degrees.

Some DEA studies account for research with published articles or citations, but these are not used in this thesis. Partly as this data was not available for Aalto BIZ individually and partly because it would broaden DEA input-output combinations and produce somewhat unreasonable assessment criteria such as citations-to-students. To compensate for research focus, the model

includes doctoral students as an intermediate variable and doctoral graduates as an output as a teaching focused proxy for research focus.

Whether DEA inputs should simply include all resource use as a single variable or split it into multiple categories is a subject of some debate. Herberholz & Wigger (2021) argue that total expenditures already capture all resource use and that splitting it into multiple variables provides limited additional diagnostic value. More generally, expanding the number of variables in the input-output set may weaken the discriminatory power of DEA results as more institutions can identify favourable weights or appear on the efficient frontier (Boussofiane et al., 1991; Dyson et al., 2001). This also increases the permissiveness of the model and reducing distinction between units. Conversely, separate input categories can represent differences in resource allocation that a single expenditure variable necessarily treats as substitutable.

In this thesis, models based on only total expenditure produced a less stable frontier and more super-efficiency problems where some units with unusual input-output combinations had disproportionate impact on efficiency scores. This does not prove that larger input sets are superior, but they can better account for heterogeneity between institutions in this sample. Consequently, resource use is split into two components: full-time equivalent staff headcount and non-staff spending in purchasing power adjusted (PPP) euros. Even when adjusting for purchasing power parity, staff wages vary between countries due to local market conditions. As a result, some researchers prefer to use staff headcounts rather than staff spending (Andersson & Sund, 2022). This normalises for varying wage levels across countries but also may introduce bias from differences in staff mix. As a result, a model using staff spending in place of staff headcount is included in the sensitivity section. Figure 4 shows the main DEA model specification used for both stationary and year-over-year benchmarking.

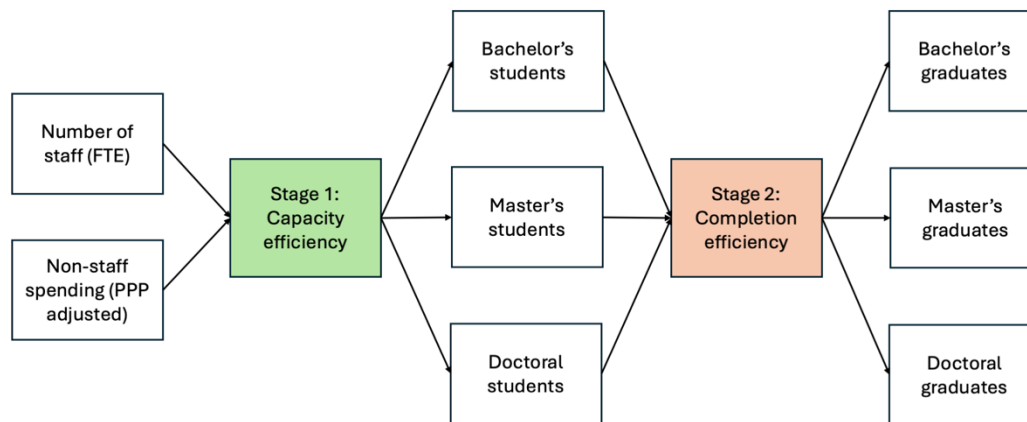


Figure 4. DEA model specification.

It is worth explicitly stating what is measured by each component of the model. Stage 1 measures educational access: the number of students who are enrolled in the university at a given time for its resource use. Importantly, the student variables measure student stocks, not annual intake or average cohort size. Thus, in addition to admission volumes, they are affected by programme duration, retention, and average completion times. Stage 2 measures throughput efficiency: how many degrees are produced relative to the size of the enrolled student body. Individual stage efficiencies are diagnostic of environment and strategy rather than strict indicators of acumen or weakness in a specific area of educational production. For performance diagnostics, overall scores, calculated as a geometric mean of the two stages are superior. Outside the methodology section, stage 1 and stage 2 efficiency are referred to as capacity efficiency and completion efficiency respectively.

The algorithm derives weights for each DMU by enveloping the data and comparing them to sampled production technologies. Therefore, a larger number and more similar DMUs leads to a larger set of observed comparable production technologies which increases the reliability of results.

There are some additional specifications which must be determined in DEA. The model orientation can be chosen between input and output oriented which determines which variables are held constant while others are radially adjusted using the most efficient production technology available among all DMUs. In input-oriented DEA, the outputs are held constant while the inputs are radially adjusted. Therefore, DMUs are efficient if they cannot produce the same outputs without using less of each resource. In output-oriented

DEA, they are efficient if they cannot produce more of each output while using the same inputs. Generally, the side with less institutional control should be held constant to most accurately model production decisions. The main model therefore adopts an input orientation in stage 1 as institutional resource use can be changed more quickly than student stocks. Total enrolment also depends on previous cohorts and graduation timelines, though resource use cannot be freely adjusted either. Especially for public universities, funding and student intake are controlled by or subject to negotiation with the state while private funding for public or private institutions is often conditional on its allocation (OECD, 2025c). In stage 2 the decision is clearer, and an output orientation should be adopted as the number of students are already given by stage 1. To model the process consistently, we must take the student counts from stage 1 as a constant without institutional control and evaluate how efficiently institutions turn those students into graduates. In the case of higher education, institutions face constraints on both sides.

In addition to choosing an orientation, DEA can be run with different assumptions on scale economies. Under constant returns to scale (CRS) outputs are assumed to change proportionally with inputs. It combines managerial efficiency and size efficiency and is appropriate when organisations do not face large constraints on their size. Variable returns to scale (VRS) relaxes the proportionality constraint and isolates pure technical efficiency regardless of size optimality. VRS constructs a locally convex frontier which permits different efficient frontiers at different scales (Banker, 1996). Universities generally have multiple frictions affecting their ability to scale such as physical capacity, regulatory constraints, friction in faculty hiring and strategic limits on student enrolment.

CRS is still useful as it is required for estimating the scale efficiency of each institution which can be used to determine their existence and direction.

$$\text{Scale efficiency} = \frac{\text{CRS efficiency}}{\text{VRS efficiency}}$$

Using a VRS model which has no scale effects will lead to both small and large DMUs having their efficiency biased upwards (Dyson et al., 2001). Additionally, VRS has a predisposition to envelop the data more closely as it uses the data itself to shape the efficient frontier, potentially leading to weakly higher efficiency scores and reduced discrimination (Dyson et al., 2001). Banker (1996) describes approaches to test returns to scale in the data. These can be used to determine whether there are variable returns to scale in addition to simply trialling both approaches.

The following VRS envelopment formulas follow the standard DEA representation in Charnes et al. (1978), with the convexity constraint associated with variable returns to scale as described in subsequent DEA literature (Banker, 1996). For stage 1 the linear programming problem is the following:

$$\begin{aligned}
 & \min_{\theta_j, z} \theta_j \\
 & \text{subject to} \\
 & \sum_{k=1}^K z_k x_{k,n} \leq \theta_j x_{j,n}, \quad n = 1, \dots, N \\
 & \sum_{k=1}^K z_k y_{k,m} \geq y_{j,m}, \quad m = 1, \dots, M \\
 & \sum_{k=1}^K z_k = 1 \\
 & z_k \geq 0, \quad k = 1, \dots, K
 \end{aligned}$$

For stage 2 the linear programming problem is:

$$\begin{aligned}
 & \max_{\phi_j, z} \phi_j \\
 & \text{subject to} \\
 & \sum_{k=1}^K z_k x_{k,n} \leq x_{j,n}, \quad n = 1, \dots, N \\
 & \sum_{k=1}^K z_k y_{k,m} \geq \phi_j y_{j,m}, \quad m = 1, \dots, M \\
 & \sum_{k=1}^K z_k = 1 \\
 & z_k \geq 0, \quad k = 1, \dots, K
 \end{aligned}$$

As the directionality of the measurement is different between stages, stage 2 scores are adjusted to be in line with the 0-1 range of stage 1 scores as:

$$E_{2j} = \frac{1}{\phi_j}$$

The overall efficiency score is then calculated as the geometric mean of the stage scores:

$$E_j^{geo} = \sqrt{E_{1j}E_{2j}}$$

Where:

- j : evaluated institution
- k : observed peer institution
- n : input index
- m : output index
- $x_{k,n}$: amount of input n used by institution k
- $y_{k,m}$: amount of output m produced by institution k
- z_k : weight assigned to peer institution k
- θ_j : input-oriented efficiency score
- ϕ_j : output-oriented expansion factor
- E_{1j} : reported Stage 1 efficiency score
- E_{2j} : reported Stage 2 efficiency score
- E_j^{geo} : reported overall efficiency score

All of the reported efficiency scores in this thesis are also bias corrected with a bootstrapping procedure in line with Simar & Wilson (2000, 2007). To account for sampling uncertainty and upward bias in DEA scores, bootstrap correction was applied separately to both stages. For each stage 1,000 bootstrap draws were generated under the original DEA specifications. The process approximates repeated sampling by generating pseudo-samples from the observed data, re-estimating the frontier comparing resulting efficiency scores across draws. These draws are then used to estimate bias and derive bias-corrected scores for each DMU which were used to calculate overall efficiency scores.

Yearly DEA measures relative efficiency by comparing DMUs against the observed frontier each year. This reveals Aalto BIZ's position relative to its peers but does not show absolute productivity change. Consequently, productivity development over time is evaluated using the Malmquist productivity index (MPI). The purpose is to determine whether changes in Aalto BIZ's relative efficiency are accompanied by improvements in its own input-output performance, rather than merely reflecting changes in the performance of the comparison group (Färe et al., 1994).

The Malmquist productivity index decomposes productivity change into two sources. Efficiency change (EC) measures movement toward or away from the best-practice frontier. Conversely, technological change (TC) measures shift in the frontier itself between periods. The decomposition distinguishes whether productivity changes arose because an institution becomes better at exploiting existing production technology (efficiency change) or because the production possibilities themselves expand or contract over time (technological change) (Färe et al., 1994). Under variable returns to scale, efficiency change can be further decomposed into two components. Pure efficiency change (PECH) reflects changes in managerial efficiency relative to the VRS frontier while scale efficiency change (SECH), reflects movement toward or away from the most productive scale size (Färe et al., 1994; Ray & Desli, 1997). The relationship is:

$$MI = EC \times TC$$

$$EC = PECH \times SECH$$

MPI was estimated between consecutive yearly observations for each institution using the same DEA specification as the stationary analysis. Stage 1 retained an input orientation and stage 2 an output orientation under VRS technology. Annual productivity changes were then aggregated using the geometric mean to obtain average productivity growth over the study period.

3.4 Robustness and limitations

The research problem and methodology establish a few key limitations which have partly been discussed in previous sections. Relative benchmarking with DEA assumes comparable production technology across DMUs, which is a necessarily imperfect assumption. This is alleviated through careful sample construction, clustering, and the two-stage model but some unobserved heterogeneity remains. Due to the relative benchmarking, the results are contingent on the sample and especially the most efficient institutions within it which determine the frontier construction that all other DMUs are benchmarked against. The frontier and results are also dependent on model specification and variable selection which is in turn informed by data availability.

The production function of HEIs is uncertain and imperfectly observed, and its measurement is limited by available data. Therefore, the efficiency scores represent a rough approximation of the production function. This could result in bias where institutions with a broader mission or more significant focus in unmeasured areas such as research output may appear less efficient. Quality is similarly left unmeasured due to its dependence on student perceptions and the lack of available data. The acquisition and integration of

informative quality metrics is an important goal whose pursuit is left to subsequent research.

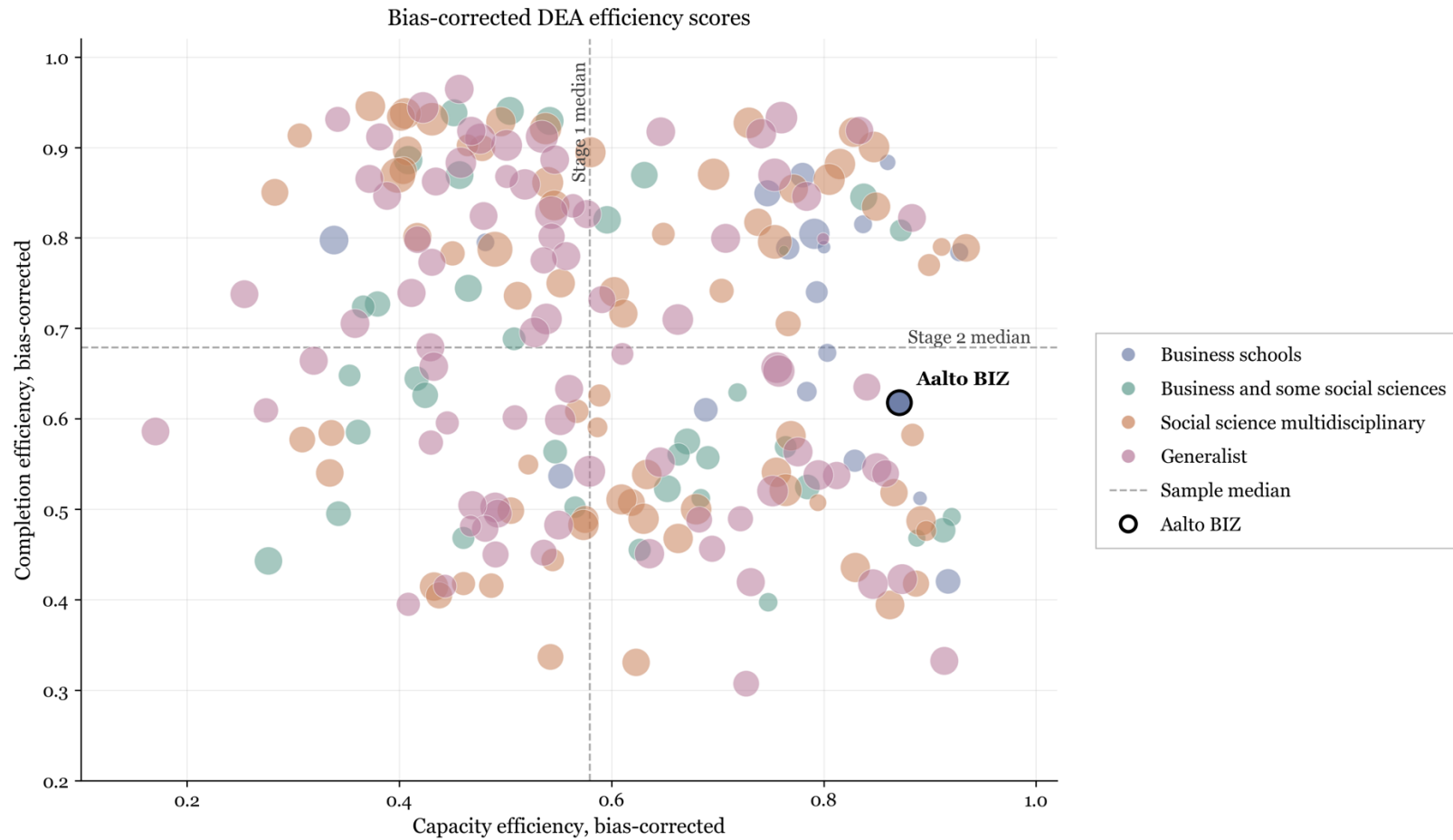
The scores are also not causal, and they do not separate inefficiency from statistical noise, measurement error, or environmental conditions. Bias correction and robustness checks improve the reliability of estimates, but the results should be interpreted as indications of rough relative positions, peer comparisons, and institutional patterns rather than indications of managerial performance or optimal resource allocations.

4 Results

This chapter presents the results of the empirical study. First the stationary DEA with averaged data for the most robust estimates of current efficiency. Second the year-over-year comparison for overall efficiency development trends for both Aalto BIZ and sampled higher education institutions overall. The results suggest that Aalto BIZ is very efficient compared to all universities, while on similar level with other business schools. Its efficiency relative to both groups has improved substantially in the 2010s.

4.1 Stationary average efficiency

Figure 5 shows a scatterplot of institutional efficiencies with the performance of Aalto BIZ and Aalto University highlighted. These are obtained by averaging the DEA variables over a 5-year period. For most institutions, including Aalto BIZ, this represents the period from 2015-2019, though a few DMUs had to use a slightly earlier time period as they were missing some data in the most recent years. Average degrees are calculated with a two-year lag relative to other variables, 2017-2021 for most institutions. The position on the graph shows the institutions' efficiency scores in capacity and completion efficiency and the size of the dot corresponds to the size of the university measured by its total PPP adjusted budget.



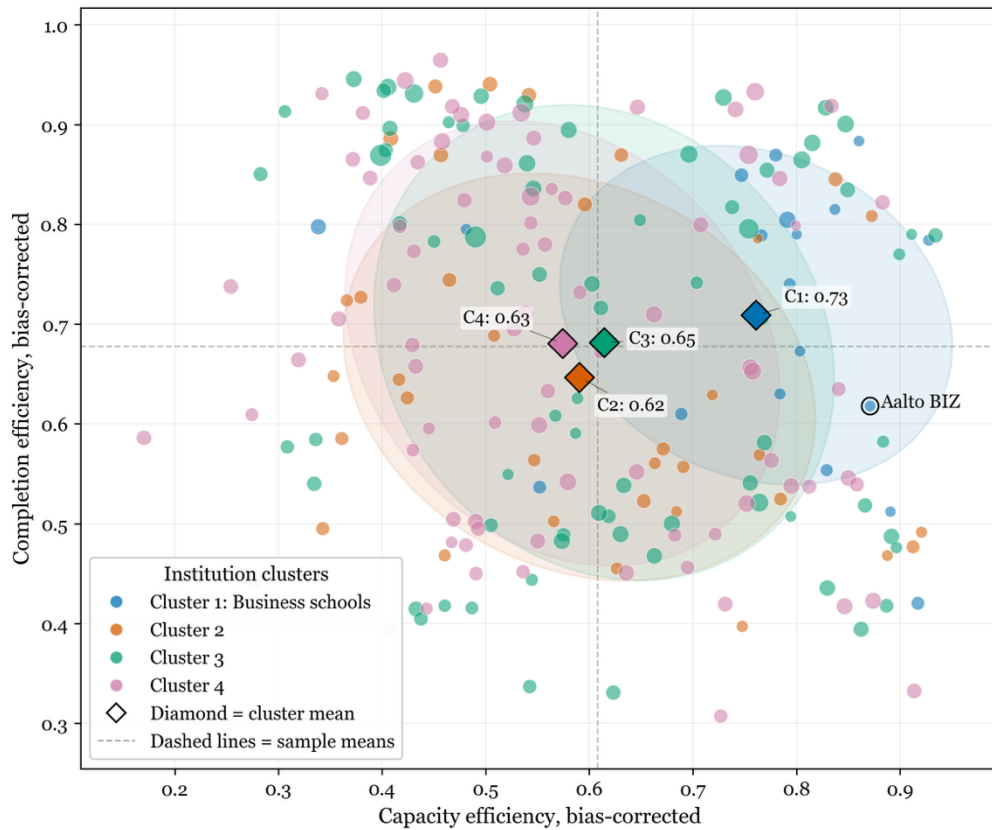
Note. Scores are bias-corrected DEA estimates. Dashed lines indicate sample medians. Marker size is proportional to log institutional budget.

Figure 5. Bias-corrected capacity and completion efficiency scores.

From Figure 5, we can observe that Aalto BIZ is extremely efficient in capacity efficiency. Its bias-corrected efficiency score is among the highest in the entire sample. Accordingly, it has a large number of enrolled students relative to its resource use. In completion efficiency Aalto BIZ is just below the median in this sample, meaning it is comparatively weak at producing degrees relative to student stocks.

We can also observe that business schools appear systematically more efficient than other institutions. They are not uniquely efficient as some institutions from all clusters achieve comparable efficiency scores in the top right quadrant of the scatterplot. The strong average performance of business schools is driven by both, a comparatively high proportion of strong performers and a very low proportion of poor performers. These differences support the clustering logic when evaluating universities with differing field mixes and suggests that the sampled business schools are comparatively homogeneous compared to the other clusters.

Other clusters have a number of institutions in each quadrant drawn by the medians (Figure 10 in the appendix), and the average efficiencies of clusters 2 to 4 are very similar as shown in Figure 6. Earlier findings that a few fields such as engineering and medicine, are particularly resource intensive, would suggest that generalist institutions are less efficient than those focused on social sciences. In this sample, business schools appear more efficient on average, but universities with medicine and engineering departments are not systematically less efficient than those focused on social sciences.



Note: Diamonds show cluster means. Labels next to diamonds report the geometric mean of mean Stage 1 and Stage 2 efficiency. Shaded ellipses summarize within-cluster dispersion and orientation. (1.25 standard deviations from cluster mean along principal directions of variation)

Figure 6. Mean cluster efficiencies and intra-cluster efficiency variation.

Table 4 shows the exact efficiency scores for Aalto BIZ, Aalto University and the medians for each cluster. While Aalto BIZ is in the 82nd percentile among all universities in overall efficiency, it is only average among business schools. Other clusters have very similar median efficiency scores but differ in capacity efficiency, leading to worse overall performance. This could reflect uncaptured resource use, for example greater research expenditures, in other clusters relative to business schools. We can also compare the efficiency of Aalto University in the ETER data to Aalto BIZ constructed from a separate internal data set. Aalto University is weaker in both capacity and completion efficiency and has a similarly skewed efficiency profile as Aalto BIZ where capacity is significantly stronger which shows no obvious incompatibility between Aalto BIZ and ETER data.

Table 4. Stationary average efficiencies for Aalto and clusters.

Institution or cluster	Capacity efficiency	Completion efficiency	Overall efficiency	Sample percentile	Distance from sample median
Aalto BIZ	0.87	0.62	0.73	82.3	0.11
Cluster 1 median	0.79	0.78	0.74	83.3	0.11
Cluster 2 median	0.58	0.61	0.61	43.6	-0.02
Cluster 3 median	0.59	0.74	0.62	47.9	0
Cluster 4 median	0.54	0.68	0.61	45.0	-0.01
Aalto University	0.73	0.42	0.55	26.3	-0.07

Cluster 1: Business schools

Cluster 2: Business and social science

Cluster 3: Broad social science focus

Cluster 4: Broad generalist

4.1.1 Discussion of results on stationary average efficiency

Capacity and completion efficiencies have a small negative Pearson correlation of -0.17. This is somewhat expected as larger enrolled student bodies lead to better capacity efficiency, but they can be caused in part by delayed completions. Institutions with longer graduation times will see larger student stocks as students remain enrolled for a longer period but also a lower number of graduations relative to that stock, leading to lower completion efficiency. Capacity efficiency shows lower minimum values while standard deviations are almost identical between the two efficiencies. A potential hypothesis is that universities have more discretion on their resource use and the number of students they admit compared to graduations as they also depend on the students' inputs. Lower minimum values on capacity efficiency may also reflect a higher proportion of unmeasured outputs, such as research contributions, especially for large institutions.

Country level differences must be evaluated against their cluster frequencies. Finnish universities appear to be somewhat below average in completion efficiency with a mean efficiency of 0.48 compared to an average of 0.60 for other countries. Average completion efficiency is also quite constant between clusters in the whole sample, so cluster mix is unlikely to bias the results. By contrast, in capacity efficiency, Finnish universities were very close to the average.

The Finnish system has historically pushed students to longer completion times with generous student support, no tuition fees, and lifetime study rights prior to 2005. As the prevalence of higher education increased, these systems have been adapted to save resources and encourage quicker completions, but they may still have an impact on the behaviour and expectations of students, employers, and universities.

Universities from the UK and Norway stand out for their low spending per graduate, primarily driven by efficient graduate throughput and low overall spending respectively. Spending in Norwegian universities is so consistently lower than other countries that it is likely caused, at least in part, by environmental differences. UK universities have systematically higher completion efficiencies. Tuition fees and strict completion timelines may explain this variation at least in part. Another country which does well is the Netherlands which also has tuition fees, although these are set by the government below market rate. Some Dutch universities have requirements for students to complete a minimum number of credits in their first year to maintain eligibility.

Other Nordic countries also do slightly better on average compared to Finnish universities. These are institutions operating in a similar environment,

their proportion of business schools is similar or lower, and they have comparable capacity efficiencies on average. Still, cultural differences may exist around students working during studies, bottlenecks especially related to the master's thesis, and with general attitudes about study completion times. Of the comparison countries, only Germany has lower completion efficiency than Finland on average.

In summary, when looking at averaged data over the period 2015-2019, Aalto BIZ is very efficient compared to other universities and around average among the small sample of specialised business schools. Its profile is skewed with very high capacity efficiency and comparatively low completion efficiency. This pattern is shared among sampled Finnish universities and contrasts against countries such as the UK and the Netherlands which have comparatively high completion efficiencies.

4.2 Aalto BIZ comparison with peer institutions

Due to the cluster size of just 19 pure business schools, it is not feasible to include doctoral students and graduates in the model as the number of variables would become too large relative to the number of DMUs. As Aalto BIZ has the third highest number of doctoral students relative to its size and is above average in doctoral graduates relative to size, their exclusion slightly lowers its efficiency relative to other business schools. Otherwise, the efficiency scores and rankings are comparable between DEA with only business schools and the business school cluster within DEA including all clusters. Business schools are significantly more efficient on average; they receive very similar efficiency scores regardless of the clustering set up as they are mostly on the frontier or their efficiency is calculated relative to the frontier consisting mainly of business schools.

The results in a smaller sample are also generally less stable and there is more super efficiency for example. Some DMUs have unique input output combinations that cannot be matched with weights from other DMUs, reducing the reliability of bootstrapping. This results in wide confidence intervals around efficiency scores and less reliable bias-corrected estimates. For these reasons, results in this section mostly focus on the scores from the full DEA run including all institutions.

Among business schools, Aalto BIZ efficiency rankings are relatively similar although slightly more extreme. Its efficiency score in capacity remains very high but its completion efficiency is ranked 17th out of 19 while in the full DEA run it is 15th out of the 19 business schools. The full business school results are presented in Table 7 in the appendix.

The clusters differ in both their resource use and student conversion to graduates. Business schools spend less per graduate, and they have more graduates per student, suggesting quicker throughput. Business-adjacent social science institutions are closer in both spending per graduate and graduation throughput while clusters 3 and 4 differ significantly and have higher spending but also a higher proportion of doctoral students.

From the cluster level differences, we can see that business schools differ in their average efficiency in this sample. In addition to better general comparability between business schools, this suggests institution level comparisons should focus on business schools for maximum comparability. The most useful peers for Aalto BIZ especially when it comes to its productivity development are other Nordic business schools which do very well in efficiency measurement as well as Hanken School of Economics which is similar in size and environment to Aalto BIZ. Table 5 shows some of the most comparable institutions to Aalto BIZ and some of their key input/output ratios.

Table 5. Key input/output ratios of Aalto BIZ and most relevant peers.

Institution	Overall efficiency	Spend / student (k€)	Spend / graduate (k€)	Bachelor grads / Bachelor stock	Master's grads / Master's stock	Doctoral share	Master's student share
Aalto BIZ	0.73	9.4	40.0	22.6%	26.3%	5.8%	45.6%
EBS University of Business and Law	0.87	16.7	51.0	31.4%	36.4%	8.7%	51.9%
Norwegian School of Economics	0.85	10.5	25.8	28.7%	52.3%	3.3%	53.1%
Stockholm School of Economics	0.83	20.1	59.7	28.7%	45.9%	7.5%	36.0%
Copenhagen Business School	0.82	8.3	25.0	27.9%	38.2%	1.4%	52.6%
Hanken School of Economics	0.68	8.0	34.1	18.5%	32.9%	4.9%	38.3%

From Table 5, we can see that Aalto BIZ has low spending per student but is slightly above average in spending per graduate. This is partly caused by its higher proportion of doctoral students, but graduation patterns also play a role. Particularly in the share of master's graduates to master's students, Aalto BIZ underperforms its peers. The practical interpretation is that students spend a significantly longer time at Aalto BIZ, especially in master's programs compared to other business schools and this is a key cause of its skewed efficiency profile. On one hand these delayed completions boost the capacity efficiency as they remain enrolled while they are less resource intensive, but they also raise student stocks without producing corresponding completions, lowering completion efficiency. Separating delayed completions from eventual dropouts is not possible with this data but additional years to completion increase institutional resource use especially around completion bottlenecks.

The cost of these delayed completions is uncertain and depends on the subject. From the institution's perspective, longer study durations likely lead to additional resource use though not in direct proportion to the duration. From the student's perspective, graduation usually boosts earnings and employability, but simultaneously employment is a common cause for delayed graduation in Finland. From a broader economic or policy perspective, graduation itself provides very marginal economic value. However, linking back to the institutional perspective, constant uniform study durations enhance planning capacity in universities and free up resources especially in bottlenecks such as thesis advisors.

We can also see that universities' resource spend per student and graduate correlate with the share of doctoral students. Despite the varying levels of spending, these universities achieve very similar efficiency scores as DEA can assign more weight for doctoral students which is justified by the greater resource demands in production.

4.3 Aalto BIZ efficiency and productivity development over time

By treating each institution year as a data point, we can calculate efficiency scores over time for institutions. The data is for the period between 2011 and 2020 reflecting the input year for the data. The yearly data presented in this section therefore consists of the resource use and student stocks from the labelled input year and graduations from input year + 2 to approximate cohort progression. Figure 7 shows the efficiency of Aalto BIZ in percentile rankings compared to the efficiency of all other institutions and the median efficiency among business schools. The efficiency of Aalto BIZ has improved consistently over the measurement period relative to other universities. This

increase has happened almost exclusively in capacity though this does not rule out improvements related to completion. If Aalto BIZ had simply increased student intake without efficiency improvement in other areas, its capacity efficiency would have risen, and completion efficiency fallen.

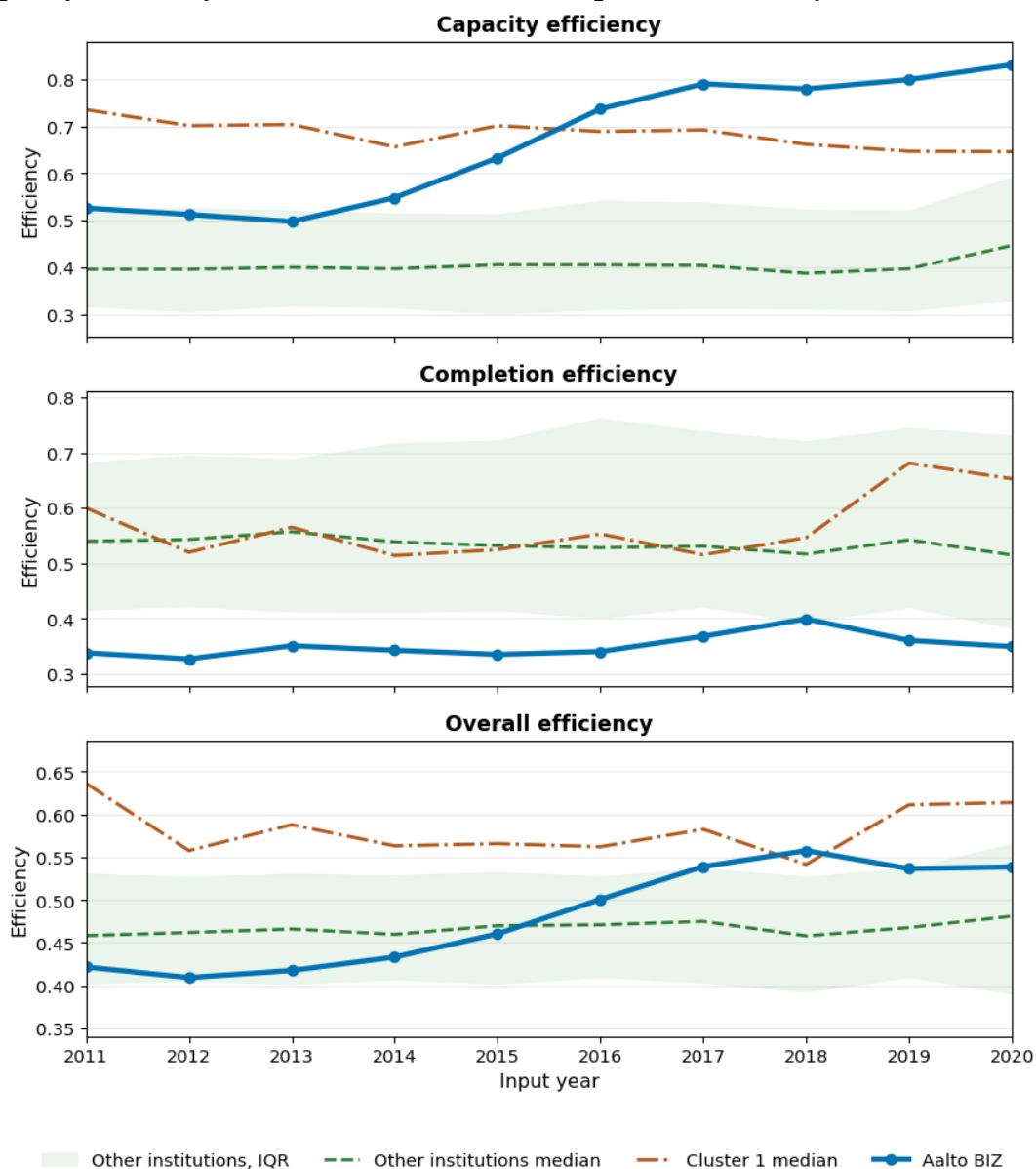


Figure 7. Development of yearly bias-corrected DEA efficiency scores for Aalto BIZ and other institutions (2011-2020).

The efficiency values in Figure 7 are slightly different compared to earlier stationary average efficiency scores as this figure treats each institution year as its own observation rather than each institution's average performance over a 5-year sample. Individual year-over-year efficiency scores are less reliable as individual years may produce outlier performance but they show how efficiencies have developed over time. In overall efficiency Aalto BIZ moved

from the bottom of the interquartile range to its top over the measurement period. In the beginning of the measurement period, Aalto BIZ was very inefficient compared to other business schools and below average among all universities. This improvement in overall efficiency is a result of its capacity efficiency improving from around 0.50 to over 0.80 while completion efficiency remained broadly stable at around 0.35-0.40. This indicates that the gains in overall efficiency resulted primarily from stronger performance in converting institutional resources into students rather than improvements in converting students into graduates.

However, this analysis is relative to the efficiency frontier observed each year. As efficiency is relative performance compared to best practices, improvements in DEA efficiency may reflect genuine productivity improvements at Aalto BIZ, weaker performance by other institutions, or both. To distinguish between these explanations, productivity development is examined using the Malmquist productivity index (MPI).

Figure 8 shows cumulative Malmquist productivity development alongside the median trajectory of each cluster. The index is normalised to 100 in 2011 with values above 100 indicating cumulative productivity growth relative to the base year.

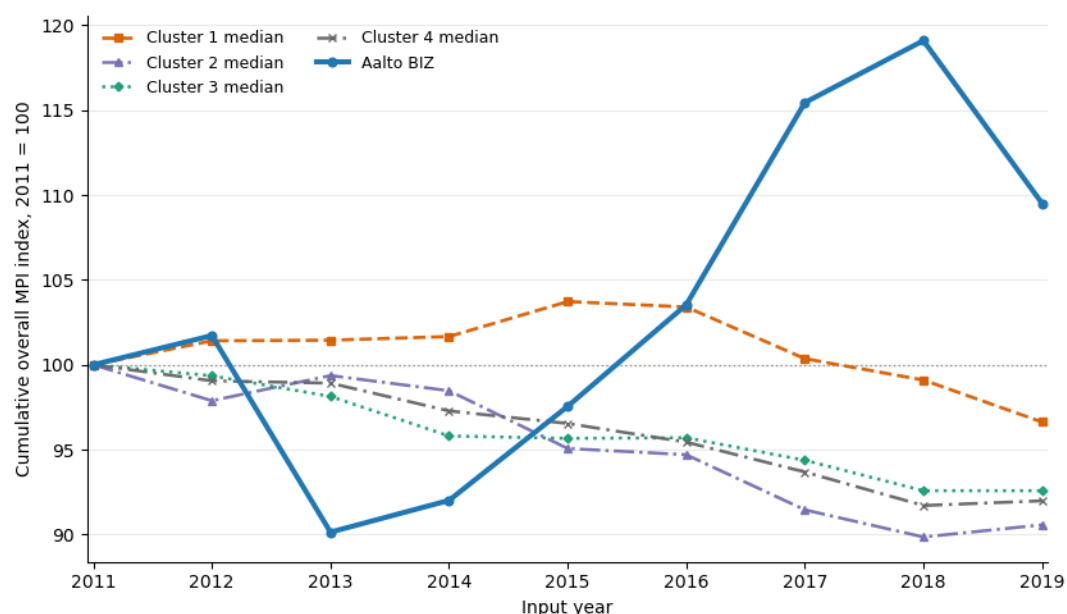


Figure 8. Aalto BIZ cumulative Malmquist productivity index compared to cluster medians.

The MPI figure allows us to see the causes of Aalto BIZ's efficiency improvement. Aalto BIZ's productivity trend diverges from the cluster medians as all clusters have seen gradual productivity decline during the measurement

period with business schools declining the least. By contrast, after an initial decline, Aalto BIZ's productivity increased continuously between 2013-2018 and its cumulative productivity growth by 2019 ranked 18th out of 191 sampled institutions. In total 47 (24.6%) improved their productivity over the measurement period. The best performing institutions by cumulative productivity development were EBS university of Business and Law, University of Bamberg, and Ulm University. Together Figure 7 and Figure 8 show that Aalto BIZ's DEA efficiency has improved over time as a result of its own productivity growth and declining productivity at peer institutions.

Table 6 presents the long-run geometric mean of the Malmquist productivity decomposition for Aalto BIZ, the business school cluster, and the full sample. It decomposes average annual productivity change between input years 2011 and 2020 into its underlying components.

Table 6. Malmquist productivity decomposition (geometric means 2011-2020).

Stage 1: Capacity	MPI	EC	TC	PECH	SECH
Aalto BIZ	1.0224	1.0411	0.9821	1.0359	1.0050
Business Schools	0.9810	0.9901	0.9908	0.9909	0.9991
Sample overall	0.9834	1.0015	0.9819	1.0120	0.9897
Stage 2: Completion	MPI	EC	TC	PECH	SECH
Aalto BIZ	1.0006	1.0316	0.9700	1.0314	1.0002
Business Schools	1.0089	1.0124	0.9966	1.0091	1.0033
Sample overall	0.9937	1.0167	0.9774	1.0074	1.0093

Note: MPI = Malmquist productivity index

EC = Efficiency change

TC = Technological change

PECH = Pure efficiency change

SECH= Scale efficiency change

The decomposition separates overall productivity change (MPI) into efficiency change (EC) and technological change (TC). Efficiency change measures whether an institution moves closer to or further from the estimated best-practice frontier whereas technological change reflects shifts in the frontier itself over time. Efficiency change can further be decomposed into pure efficiency change (PECH), representing improvements in how efficiently an institution operates given its size, and scale efficiency change (SECH), representing changes associated with distance to the most productive scale. Values above one indicate improvement while values below one show deterioration. This decomposition allows productivity growth to be attributed to specific sources.

In capacity, Aalto BIZ recorded a productivity index of 1.022, corresponding to approximately 2.2% average annual productivity growth. This productivity growth resulted from positive efficiency change ($EC = 1.041$), which more than offset negative technological change ($TC = 0.982$). The efficiency improvement was driven almost entirely by pure efficiency change ($PECH = 1.036$) while scale efficiency remained close to unity ($SECH = 1.005$). Aalto BIZ achieved notable productivity growth in capacity while other cluster averages and the overall sample experienced declining productivity.

In completion, Aalto BIZ again recorded positive efficiency change ($EC = 1.032$), driven almost entirely by pure efficiency change ($PECH = 1.031$) while scale efficiency remained effectively unchanged ($SECH = 1.000$). However, stronger negative technological change ($TC = 0.970$) largely offset the efficiency gains resulting in an overall productivity index close to unity ($MPI = 1.001$). Similar patterns were observed across the broader sample, where positive efficiency change was counteracted by negative technological change resulting in no or negative productivity growth.

Productivity ratios provide descriptive context for the factors underlying Aalto BIZ's DEA efficiency and productivity results and due to data availability for Aalto BIZ allow for measuring more recent productivity change. Figure 9 presents three partial productivity indicators for Aalto BIZ: students per total spending, graduates per student, and graduates per total spending for each input year from 2011 to 2022. The ratios are expressed as percentage changes relative to the 2011 baseline. They use similar input-output combinations as the DEA efficiencies but lack the weights optimised by the model and therefore value all types of degrees equivalently for example. These ratios therefore complement the DEA results and provide insight into Aalto BIZ's productivity development relative to its own baseline.

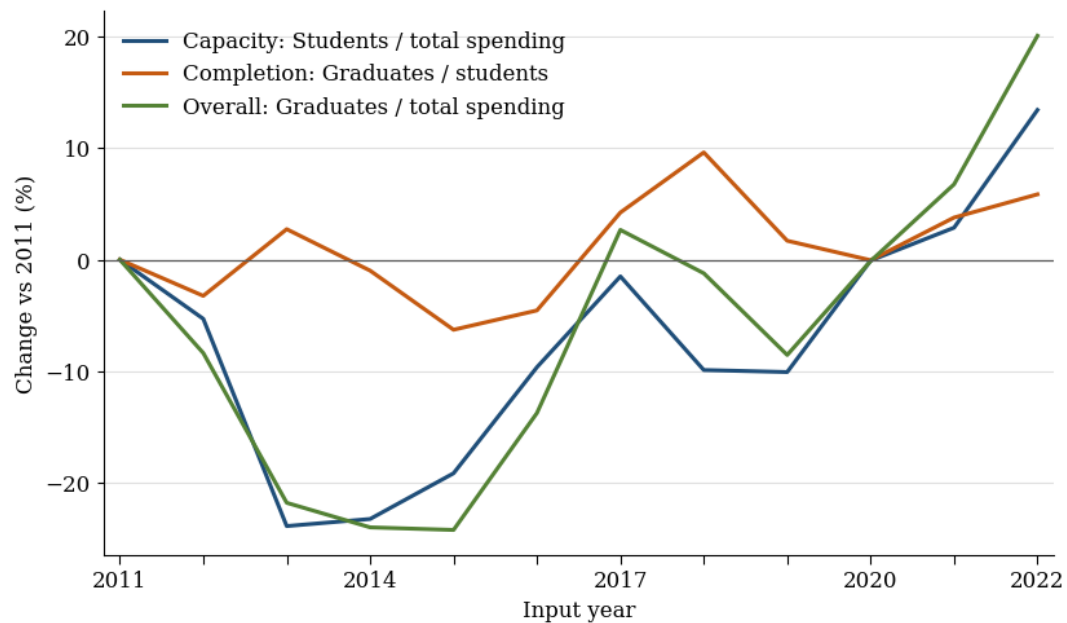


Figure 9: Aalto BIZ key productivity ratio change 2011-2022

Aalto BIZ's productivity ratios here are consistent with Figure 8 with an initial dip in productivity followed by fairly constant improvement. Calculating the ratios for Aalto BIZ exclusively also allows a longer measurement period as it is the only institution with data beyond 2021. Therefore, the final year of this plot is input year 2022 which includes graduation outcomes from 2024. We can see that beyond input year 2019 where data on other institutions ends, the performance of Aalto BIZ further improves and its ratios of graduates relative to spending in input year 2022 is highest in the measurement period, 20% greater compared to its 2011 level.

Exact values for the key productivity variables are presented in Table 8 in the appendix. Overall, we can see that Aalto BIZ's resource use measured in total spending is very similar in input years 2011 and 2022. Over the same time period the number of enrolled master's students increased while bachelor's and doctoral student numbers are similar. This is accompanied by an improvement in graduations relative to enrolled students for bachelor's and doctoral students and a decline for master's students. Aalto BIZ's resource use has held constant while it has increased capacity in master's programs and improved completions in bachelor's and doctoral programs. As Aalto BIZ's productivity is at its highest point in the input period from 2020-2022, the DEA results may understate Aalto BIZ's current performance and trajectory.

4.4 Sensitivity

Testing multiple model specifications is standard practice for DEA. Models were run with different sets of variables and samples including different lags for degrees, variations on input and output variables, and with different combinations of clusters. The differences between these model runs were generally quite small. Different lag structures created particularly little difference, presumably as student stocks are accumulated over multiple years which smooths yearly variation. Different input-output combinations did not change results much except for causing new sets of super-efficient units. After those were removed, the remaining efficiency estimates, especially for Aalto BIZ, were largely consistent across model specifications.

Generally, among super-efficient units, those whose observed production function was infeasible when using production technology from other institutions were the only institutions removed for each model run. Removing these DMUs from the DEA analysis was necessary as their inclusion would compress other DMUs' scores and disproportionately impact DMUs of similar size, pushing their efficiency down. They would also cause efficiency estimates of DMUs with similar production technology to be biased downward significantly.

Different lags have very small implications on results. Lags of 0, 1, 2, and 3 years were tried and results were barely impacted. DMUs which experienced larger than average changes in size were most impacted by differences in lags but even their differences were marginal. Size changes are generally quite small year-over-year and their impact is smoothed by the multi-year nature of the production process.

The two-stage model could theoretically distort efficiency scores when they are obtained as a combination of the stage scores. This could especially impact institutions whose efficiency is skewed towards one stage such as Aalto BIZ. The efficiency scores from a one-stage model were highly correlated with the two-stage mode. Aalto BIZ's efficiency declined slightly under this specification as enrolled students were not considered, but the decline was small, roughly 0.05 depending on exact specification.

When doctoral students and graduates are not included, the gap between business schools and other HEIs increases as business schools have fewer doctoral students relative to their student population or resource use. When benchmarking teaching efficiency, a model with fewer variables favours small institutions focused on teaching as bigger generalist institutions are likely to have comparatively more resource use in various areas which are left unmeasured by a compact model.

Overall, the sensitivity analyses indicate that the substantive conclusions of the study are robust to reasonable modelling choices. Alternative lag structures, variable specifications, and sample compositions produced only minor changes in efficiency estimates. They also did not alter the interpretation of Aalto BIZ's relative performance or the broader differences observed between institutional groups. The largest effects arose from the treatment of super-efficient institutions and the inclusion or exclusion of doctoral education, but these primarily affected the comparability of peer groups rather than the conclusions of the analysis.

4.5 Summary of findings

To summarise the results and answer the research questions, Aalto BIZ is very efficient compared to other European universities, but closer to average when compared with the narrower group of small, specialised business schools mostly focused on teaching. The selected specialised business schools were more efficient on average compared to broader universities and represent a more appropriate benchmarking group.

The most relevant peers for Aalto BIZ in efficiency comparisons are other small European business schools which systematically have a teaching focus and are highly efficient at teaching compared to larger universities with a broader field mix. Additionally, as environment matters particularly for completions, other Nordic business schools are the most relevant benchmark.

The year-over-year results suggest that Aalto BIZ's relative teaching efficiency improved through the observed period mainly through an increase in capacity efficiency. Aalto BIZ has been able to increase the number of students relative to resource use while its graduation throughput has remained steady. During the same window, the average efficiency of universities in all clusters declined. Few universities were able to improve their efficiency over the measurement period.

In summary, Aalto BIZ is comparable to other, especially Nordic business schools which have high intake of students given their resources even despite their smaller size and lack of potential scale advantages in this stage. Efficiency could be improved by increasing graduation rates and decreasing graduation times as this phase of the production process shows lacking efficiency for Aalto BIZ, particularly in master's programs. These characteristics are likely caused in part by the environment due to the flexible nature of the Finnish higher education system and improving completion efficiency is not fully in the control of Aalto BIZ.

5 Discussion

This chapter discusses the implications of the results as well as their connection to earlier literature. It is split into three subchapters, the theoretical contributions of this thesis, its managerial implications as well as limitations and future research directions.

5.1 Theoretical contributions

The results of this thesis are largely consistent with the higher education efficiency literature. As in previous studies, efficiency varied greatly between institutions, measured efficiency depended on the choice of peer group, and institutional heterogeneity complicated direct comparisons (De Witte & López-Torres, 2017; Johnes, 2006). There are also new findings, and its theoretical contributions are fourfold.

First, this thesis develops a novel conceptual service operations taxonomy (Figure 1) by synthesising and extending the work of Chase (1978), Grönroos (1984), and Sampson & Froehle (2006). It proposes classifying service operations by their mode of customer participation. This classification in turn impacts service design, standardisation, efficiency potential, and quality evaluation. This service operations taxonomy provides a theoretical justification for a two-stage model with students as intermediate products for measuring teaching efficiency.

Second, it applies this framework to higher education and provides a theoretical basis for modelling the enrolled student stock as an intermediate variable between institutional resources and completed degrees. Previous higher education network DEA studies have modelled enrolled students as intermediate measures in a sequential production process, for example Kao (2009). Broader reviews note that students have alternatively been treated as inputs, outputs, or intermediate variables (De Witte & López-Torres, 2017; Johnes, 2006). These studies generally justify such modelling choices operationally, reflecting the sequential structure of educational production. This thesis provides a theoretical explanation as it shows students are not passive outputs of resource use, nor are they conventional production inputs. The customer-supplier-object triad extends the work of Sampson (2000) on customer-supplier duality. Treating enrolled students as an intermediate stage in a two-stage model therefore reflects the underlying co-production structure more closely than a one-stage model from institutional resources directly to graduates. The empirical results for Aalto BIZ demonstrate the value of this distinction: its strong capacity efficiency performance and weaker completion

efficiency would be obscured by an aggregate resources-to-graduates measure.

Third, the empirical findings show that environmental heterogeneity operates through multiple dimensions of educational production: field mix affects the comparability of institutional resource requirements, while national rules, practices, and student behaviour appear to affect completion efficiency in particular. Previous studies have argued that estimated efficiency is influenced by differences in disciplinary composition, institutional mission, scale, and operating environment, meaning institutions should be compared against peers with similar production technologies (Agasisti & Bonomi, 2014; Bruni et al., 2020; Dyson et al., 2001; Herberholz & Wigger, 2021). The results of this thesis are consistent with this view. Business schools appeared substantially more efficient than broader universities in the pooled sample. Aalto BIZ appeared highly efficient against the full sample but had average efficiency among specialised business schools.

The thesis employed clustering to accommodate institutional heterogeneity. Average cluster differences were small outside specialised business schools which were more efficient on average (Table 4, Figure 10). Earlier research has found average efficiency differences between fields, with medicine and engineering appearing less efficient compared to social sciences (Agasisti & Bonomi, 2014; Herberholz & Wigger, 2021). Similar differences were not observed between broader institutional clusters in this thesis. By contrast, specialised business schools constitute relatively homogenous production systems and are benchmarked without aggregation across fundamentally different disciplines. Their higher average efficiency is therefore consistent with earlier evidence of disciplinary differences, while suggesting that such differences may be difficult to identify when benchmarking is conducted at the level of multidisciplinary universities. This reinforces the importance of constructing peer groups around comparable production systems.

Previous research has hypothesised that field differences could arise from different resource requirements in teaching and research. This would lead to increased resource use and a corresponding lower capacity efficiency. Larger and more heterogeneous institutions also have broader student variation and organisational complexity which could reduce completion efficiency (Lazear, 2001; Winston & Zimmerman, 2004; Agasisti & Johnes, 2015). One explanation for the relatively small differences between the multidisciplinary clusters 2-4 is that they were measured at the university level. Aggregation at the university level masks efficiency differences, as institutional profiles are relatively comparable. Detecting field composition effects among multidisciplinary universities would likely require a decomposition approaches such as Agasisti & Bonomi (2014). Larger generalist universities may also partially

offset higher resource requirements through economies of scale and shared institutional resources.

Fourth, the results also suggest that different sources of heterogeneity have asymmetric impacts on different stages of educational production. The efficiency advantage of business schools was more pronounced in capacity efficiency (Table 4, Figure 11). One explanation is that disciplinary composition primarily affects the resources required to support a given student population, whereas completion increasingly depends on processes occurring after enrolment. Previous research has shown that funding systems, regulation, tuition arrangements, institutional incentives, and national characteristics influence feasible input-output combinations and measured efficiency (Agasisti & Johnes, 2015; Andersson & Sund, 2022; Marginson, 2008; Parteka & Wolszczak-Derlacz, 2013). The observed weaker completion efficiency of Finnish institutions and the explanatory power of country indicators are consistent with this literature. Country differences were considerably more pronounced in completion efficiency and could be linked to feasible causes such as tuition fees. This suggests that national environments may influence the conversion of enrolled students into graduates more strongly than the conversion of institutional resources into educational capacity.

This interpretation is consistent with service operations perspective adopted in this thesis. Sampson & Froehle (2006) argue that service performance becomes increasingly dependent on the customer, while Frei (2006) shows that customer-induced variation constrains provider control. Within the two-stage model applied in this thesis, institutional resources dominate capacity efficiency whereas completion efficiency depends increasingly on student participation and behaviours that are shaped by both institutional practices and the wider environment (Hanushek, 1979). These findings suggest that environmental heterogeneity is multidimensional, with differing effects on different stages of the education production process. This provides a basis for distinguishing effects of managerial action from those of the environment.

The longitudinal results are partly consistent with earlier research reporting limited or uncertain productivity growth in higher education (Johnes, 2008; Parteka & Wolszczak-Derlacz, 2013). Cluster-level performance was quite stable, on average slightly declining. This supports the view that universities are slow-moving and institutionally constrained organisations. At the same time, Aalto BIZ improved substantially relative to the sector. Despite sectoral challenges, meaningful institution-level productivity improvements can occur even where the broader frontier changes are small or even declining.

5.2 Managerial implications

The results provide several implications for how Aalto BIZ and comparable higher education institutions can manage teaching efficiency within existing resource constraints. Aalto BIZ supports a comparatively large student population relative to its resource use, and its capacity efficiency improved consistently over the measurement period. At the same time, completion efficiency remained comparatively weaker, particularly at the master's level. Similar efficiency profiles were common among Finnish universities, suggesting they are affected by environmental factors such as student behaviour and incentives in addition to institutional processes. Consequently, the following suggestions take into account the operating environment and focus on areas where university management can most plausibly influence teaching efficiency while recognising the limits of managerial action.

The results suggest four primary recommended actions for Aalto BIZ leadership:

1. Focus managerial attention on completion efficiency
2. Target institutional support toward master's progression
3. Evaluate student intake jointly with downstream completion capacity
4. Monitor efficiency systematically alongside educational quality

First, managerial attention should be focused on completion efficiency. Aalto BIZ already has strong enrolment relative to resource use, but it performs poorly in completion. Without corresponding improvements in student throughput, additional enrolment increases may accumulate larger student stocks rather than a proportionally higher number of graduates. Consequently, completion efficiency becomes an increasingly important managerial objective as intake expands to preserve or further improve overall efficiency.

Second, the focus on completion improvement should be applied especially at the master's level. The graduations relative to student enrolment (Table 5) indicate that Aalto BIZ's weaker throughput is concentrated in master's programs. This result points towards master's thesis completion, programme pacing, and late-stage student support as the most relevant targets to improve on-time graduations and efficiency. In practice this could imply more active supervisor involvement, better integration into studies with a course aimed at kickstarting the thesis process, or more explicit planned deliverables around thesis milestones. Improving completion timelines is especially relevant if Aalto BIZ continues to increase student intake without corresponding increases in resource use. Previous research similarly argues that institutional practices including academic support, supervision, and

programme organisation can influence persistence and successful completion (Tinto, 1975; Yorke, 1999), while effective master's thesis supervision has been identified as an important component of timely completion (Grohnert et al., 2024).

Without process improvements further capacity gains could be offset by weakening completion performance. The observed student enrolment growth has also occurred specifically in master's programs where selectivity is lesser compared to bachelor's or doctoral programs at Aalto BIZ. Concurrently, master's programs have seen a slight reduction in graduations relative to student stock which has been compensated by improved completion performance in other programs. Intake expansion may bring students with additional or different support needs and addressing these can increase completion rates (Bound et al., 2010). Aalto BIZ has increased intake while improving its efficiency, but sustained improvements require continued attention on completion efficiency and educational quality.

Third, student intake should be managed as one component of a broader student lifecycle. Intake planning should consider immediate teaching capacity as well as ability to support students through progression, supervision, and ultimately graduation. Identifying student needs and bottlenecks in the production process should be done to analyse which phases have particularly high risk of losing student engagement. Considering the results of this thesis, the most immediate extension would be a detailed analysis of master's level progression and thesis-completion process.

Fourth, beyond the specific findings regarding institutional efficiency, this thesis demonstrates the value of systematic efficiency monitoring. Existing management information captures aspects of educational performance but continuous monitoring of these indicators as well as their integration into measures of educational efficiency could improve strategic management. Capturing all of capacity, progression, completion, and quality provides a comprehensive framework for educational efficiency and allows for observing strengths and weaknesses. Existing reporting already includes extensive information on these, but they are generally interpreted separately. The results suggest that management would benefit from periodically integrating these measures to ensure changes in individual variables represent overall improvements rather than shifts in different stages of the education process.

Aalto BIZ exhibited positive productivity development in the Malmquist productivity index decomposition through efficiency change in both production stages. This was followed with the observation that productivity ratios had continued improving further after the DEA period between 2019 and 2022. However, technological change was negative in both stages,

particularly in completion, largely offsetting these efficiency gains. In practice, this shows that Aalto BIZ mostly became relatively more efficient than comparable institutions without experiencing significant long-run productivity growth. These findings are consistent with the service operations perspective developed throughout the thesis. Higher education remains a labour-intensive, high customer participation service in which productivity improvements depend primarily on better organisation of existing processes rather than fundamental changes in production technology.

Aalto BIZ retains some scope to improve efficiency through catch-up relative to the most efficient business schools. However, when approaching the frontier, incremental improvements in processes become increasingly difficult to realise. Sustained productivity growth therefore depends increasingly on innovations capable of expanding the production possibilities. In higher education, these innovations are unlikely to consist solely of labour substitution or automation. Instead, they are more likely to arise through changes that improve the joint productivity of staff and students. Historically, examples include curriculum redesign and modular programme structures that reduce unnecessary delays but maintain learning outcomes. In the future, advances in generative AI could expand the educational production frontier by making personalised feedback, assessment support, and student guidance feasible for a much larger number of students. Because education is co-produced with students, the technical capability of a new production technology does not necessarily translate into realised productivity gains. Its effectiveness depends on how it changes student participation, learning, and engagement in practice. Monitoring educational quality alongside efficiency therefore becomes essential for distinguishing genuinely successful innovations from technologies whose potential is undermined by poor implementation or weak adoption.

5.3 Limitations and future research

The first limitation of this thesis is that the model is a limited view of the university production function. The present model focuses on observable teaching throughput by relating institutional resources to student stocks and graduates. It does not directly measure learning gains, graduate quality, student experience, labour-market outcomes, or broader social value. Research contributions are minimally accounted for through doctoral students and graduates. The results should therefore be interpreted as measures of teaching efficiency rather than overall university performance. Teaching-focused institutions may perform relatively well because a larger share of their activities is represented in the production model. The higher average efficiency of business schools may therefore reflect both operational performance and better alignment between their activities and the selected variables.

Although multiple model specifications were tested and produced similar results, DEA estimates remain conditional on the chosen model specifications. Including research output could improve comparability, but it would also create additional problems of attribution, timing, field normalisation, and weighting. Teaching and research are jointly produced using partly shared resources, which makes clean separation difficult (Johnes & Johnes, 2009; Johnes & Yu, 2008). Moreover, adding variables does not necessarily improve a DEA model. Additional dimensions reduce discrimination and increase the number of institutions that can appear efficient, particularly in small samples (Dyson et al., 2001). Future work should therefore compare teaching-only and joint-production models rather than assume that the most comprehensive specification is automatically the most valid.

Similarly educational quality is clearly of interest to institutions, and its absence is a limitation on the model. Existing quality indicators such as student evaluations, labour-market outcomes, and composite quality measures, such as rankings, suffer from limited availability and questionable validity, and their exclusion is a conscious choice to improve the reliability of results. The development and integration of quality measures to efficiency benchmarking is left to future research. While integrating quality indicators to efficiency research is a desirable goal for future research, focusing on several complementary indicators of quality alongside efficiency research results may be a more realistic path. Relevant measures could include learning outcomes, completion patterns, and student experience. The goal should be to test the relationship between throughput and quality to understand the effects of efficiency improvements on quality dimensions.

A second limitation concerns the treatment of student stocks as intermediate products. In addition to admissions and teaching capacity, student stocks also reflect retention, inactivity, programme duration, and delayed completion. Consequently, longer study times can simultaneously raise capacity efficiency by increasing enrolled student stocks and reduce completion efficiency by weakening the ratio of graduates to students. The two-stage model reveals differing efficiency profiles, but aggregate institution-level data cannot distinguish the exact cause. Completion inefficiency may be caused for example by slow active study, delayed thesis completion, temporary inactivity or high dropout rates.

Consequently, future research should attempt to move from aggregate stocks toward cohort-based measures that follow students through the educational process, allowing more accurate attribution of progression, completion, and institutional performance (Jain & Gulati, 2025; Johnes, 2006). These include time-to-degree distributions, annual credit accumulation, active and inactive enrolment, and graduation trends which would better identify causes and costs of inefficiency.

A third limitation is the heterogeneity of institutional and national environments. DEA assumes comparable production technologies, yet the results show that clustering is valuable but insufficient in removing these differences. This means that distance from the frontier cannot be solely attributed to managerial inefficiency. A university may therefore appear efficient or inefficient partly because of its student body, regulatory environment, field mix, or completion norms rather than because of managerial performance alone.

A fourth limitation concerns data availability and harmonisation. ETER offers broad cross-national coverage, but missing observations and data aggregation required sample restriction and may introduce minor measurement error. As data for Aalto BIZ is sourced internally, some differences between accounting boundaries, shared university services, and personnel definitions may remain.

Finally, efficiency benchmarking should be regarded as one component of decision support. Higher education benchmarking has long been contentious, particularly where funding systems reward a limited set of output indicators (Elton, 2004; Jongbloed et al., 2018; Pruvot & Estermann, 2022; Vakkuri & Meklin, 1998). Researchers have criticised university “league tables”, arguing that relative performance measures can encourage oversimplified comparisons and distort organisational behaviour when they become targets rather than diagnostics (Dill & Soo, 2005; Espeland & Sauder, 2007; Morrison et al., 1995; Sauder & Espeland, 2009). Although these critiques primarily concern rankings, the underlying concerns also apply to efficiency benchmarking. Efficiency estimates should therefore complement broader assessments incorporating educational quality, research performance, institutional mission, and contextual factors. Used in this way, benchmarking is best understood as a tool for identifying potential areas for investigation and improvement rather than providing definitive judgements of institutional performance.

References

- Abbott, M., & Doucouliagos, C. (2003). The efficiency of Australian universities: a data envelopment analysis. *Economics of Education Review*, 22, 89–97. [https://doi.org/10.1016/S0272-7757\(01\)00068-1](https://doi.org/10.1016/S0272-7757(01)00068-1)
- Abbott, M., & Doucouliagos, C. (2009). Competition and efficiency: overseas students and technical efficiency in Australian and New Zealand universities. *Education Economics*, 17(1), 31–57. <https://doi.org/10.1080/09645290701773433>
- Abramovitz, M. (1956). *Resource and Output Trends in the United States since 1870*.
- Agasisti, T., & Bonomi, F. (2014). Benchmarking universities' efficiency indicators in the presence of internal heterogeneity. *Studies in Higher Education*, 39(7), 1237–1255. <https://doi.org/10.1080/03075079.2013.801423>
- Agasisti, T., & Johnes, G. (2009). Beyond frontiers: comparing the efficiency of higher education decision-making units across more than one country. *Education Economics*, 17(1), 59–79. <https://doi.org/10.1080/09645290701523291>
- Agasisti, T., & Johnes, G. (2015). Efficiency, costs, rankings and heterogeneity: the case of US higher education. *Studies in Higher Education*, 40(1), 60–82. <https://doi.org/10.1080/03075079.2013.818644>
- Agasisti, T., & Pérez-Esparrells, C. (2010). Comparing efficiency in a cross-country perspective: The case of Italian and Spanish state universities. *Higher Education*, 59(1), 85–103. <https://doi.org/10.1007/S10734-009-9235-8/TABLES/10>
- Aigner, D., Lovell, C. A. K., & Schmidt, P. (1977). Formulation and estimation of stochastic frontier production function models. *Journal of Econometrics*, 6(1), 21–37. [https://doi.org/10.1016/0304-4076\(77\)90052-5](https://doi.org/10.1016/0304-4076(77)90052-5)
- Andersson, C., & Sund, K. (2022). Technical Efficiency and Productivity of Higher Education Institutions in the Nordic Countries. *International Journal of Public Administration*, 45(2), 107–120. <https://doi.org/10.1080/01900692.2020.1868508>
- Angrist, J. D., Hull, P. D., Pathak, P. A., & Walters, C. R. (2017). Leveraging Lotteries for School Value-Added: Testing and Estimation. *The Quarterly Journal of Economics*, 132(2), 871–919. <https://doi.org/10.1093/QJE/QJX001>
- Banker, R. D. (1996). Hypothesis Tests Using Data Envelopment Analysis. *Source: Journal of Productivity Analysis*, 7(3), 139–159. <https://www.jstor.org/stable/41770797>
- Baumol, W. J. (1967). Macroeconomics of Unbalanced Growth: The Anatomy of Urban Crisis. *The American Economic Review*, 57(3), 415–426.
- Belfield, C. R., & Fielding, A. (2001). Measuring the relationship between resources and outcomes in higher education in the UK. *Economics of Education Review*, 20, 589–602. www.elsevier.com/locate/econedurev
- Bound, J., Lovenheim, M. F., & Turner, S. (2010). Why Have College Completion Rates Declined? An Analysis of Changing Student Preparation and Collegiate

- Resources. *American Economic Journal: Applied Economics*, 2(3), 129–157.
<https://doi.org/10.1257/APP.2.3.129>
- Boussofiane, A., Dyson, R. G., & Thanassoulis, E. (1991). Applied data envelopment analysis. *European Journal of Operational Research*, 52(1), 1–15.
[https://doi.org/10.1016/0377-2217\(91\)90331-O](https://doi.org/10.1016/0377-2217(91)90331-O)
- Bruinsma, M., & Jansen, E. P. W. A. (2007). Educational productivity in higher education: An examination of part of the Walberg educational productivity model. *School Effectiveness and School Improvement*, 18(1), 45–65.
<https://doi.org/10.1080/09243450600797711>
- Bruni, R., Catalano, G., Daraio, C., Gregori, M., & Moed, H. F. (2020). Studying the heterogeneity of European higher education institutions. *Scientometrics*, 125(2), 1117–1144. <https://doi.org/10.1007/S11192-020-03717-W/TABLES/10>
- Carrell, S. E., Sacerdote, B. I., & West, J. E. (2013). From Natural Variation to Optimal Policy? The Importance of Endogenous Peer Group Formation. *Econometrica*, 81(3), 855–882. <https://doi.org/10.3982/ECTA10168>
- Charnes, A., Cooper, W. W., & Rhodes, E. (1978). Measuring the efficiency of decision making units. *European Journal of Operational Research*, 2(6), 429–444.
[https://doi.org/10.1016/0377-2217\(78\)90138-8](https://doi.org/10.1016/0377-2217(78)90138-8)
- Chase, R. B. (1978). Where Does the Customer Fit in a Service Operation? *Harvard Business Review*, 56(6), 137–142.
- Chase, R. B. (2010). *Handbook of service science*. https://doi.org/10.1007/978-1-4419-1628-0_2
- Clark, C. (1940). *The conditions of economic progress*.
- Clauset, A., Shalizi, C. R., & Newman, M. E. J. (2009). Power-law distributions in empirical data. *SIAM Review*, 51(4), 661–703.
<https://doi.org/10.1137/070710111>
- Cossani, G., Codoceo, L., Cáceres, H., & Tabilo, J. (2022). Technical efficiency in Chile's higher education system: A comparison of rankings and accreditation. *Evaluation and Program Planning*, 92, 102058.
<https://doi.org/10.1016.2022.102058>
- Darby, M. R., & Karni, E. (1973). Free Competition and the Optimal Amount of Fraud. *Source: Journal of Law and Economics*, 16(1), 67–88.
- De Witte, K., & López-Torres, L. (2017). Efficiency in education: a review of literature and a way forward. *Journal of the Operational Research Society*, 68(4), 339–363. <https://doi.org/10.1057/JORS.2015.92>
- Diewert, W. E. (1992). Fisher ideal output, input, and productivity indexes revisited. *Journal of Productivity Analysis*, 3(3), 211–248.
<https://doi.org/10.1007/BF00158354/METRICS>
- Dill, D. D., & Soo, M. (2005). Academic quality, league tables, and public policy: A cross-national analysis of university ranking systems. *Higher Education*, 49(4), 495–533. <https://doi.org/10.1007/S10734-004-1746-8/METRICS>
- Dochy, F., Segers, M., & Buehl, M. M. (1999). The Relation Between Assessment Practices and Outcomes of Studies: The Case of Research on Prior

- Knowledge. *Review of Educational Research*, 69(2), 145–186.
<https://doi.org/10.3102/00346543069002145>
- Doherty, G. D. (2008). On quality in education. *Quality Assurance in Education*, 16(3), 255–265. <https://doi.org/10.1108/09684880810886268>
- Dyson, R. G., Allen, R., Camanho, A. S., Podinovski, V. V., Sarrico, C. S., & Shale, E. A. (2001). Pitfalls and protocols in DEA. *European Journal of Operational Research*, 132(2), 245–259. [https://doi.org/10.1016/S0377-2217\(00\)00149-1](https://doi.org/10.1016/S0377-2217(00)00149-1)
- Elton, L. (2004). Goodhart’s Law and Performance Indicators in Higher Education. *Evaluation & Research in Education*, 18(1–2), 120–128.
<https://doi.org/10.1080/09500790408668312>
- Espeland, W. N., & Sauder, M. (2007). Rankings and reactivity: How public measures recreate social worlds. *American Journal of Sociology*, 113(1), 1–40.
<https://doi.org/10.1086/517897>
- ETER. (2025). *Handbook for the European Tertiary Education Register data collection*. <https://doi.org/10.2797/5115139>
- Färe, R., Grosskopf, S., Norris, M., & Zhang, Z. (1994). *Productivity Growth, Technical Progress, and Efficiency Change in Industrialized Countries*. 84(1), 66–83.
- Farrell, M. J. (1957). The Measurement of Productive Efficiency. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 120(3), 253–281.
<https://doi.org/10.2307/2343100>
- Fisher, A. G. B. (1939). Production, Primary, Secondary And Tertiary. *The Economic Record*, 15(1), 24–38. <https://doi.org/10.1111/J.1475-4932.1939.TB01015.X>
- Frei, F. X. (2006). Breaking the trade-off between efficiency and service. *Harvard Business Review*, 84(11), 93–101, 156. <https://europepmc.org/article/med/17131566>
- Fried, H. O., Knox Lovell, C. A., & Schmidt, S. S. (2008). The Measurement of Productive Efficiency and Productivity Change. *The Measurement of Productive Efficiency and Productivity Change*, 1–641. <https://doi.org/10.1093/ACPROF:OSO/9780195183528.001.0001>
- García-Aracil, A., & Palomares-Montero, D. (2010). Examining benchmark indicator systems for the evaluation of higher education institutions. *Higher Education*, 60(2), 217–234. <https://doi.org/10.1007/S10734-009-9296-8/TABLES/5>
- Geuna, A., & Martin, B. R. (2003). University research evaluation and funding: An international comparison. *Minerva*, 41(4), 277–304.
<https://doi.org/10.1023/B:MINE.00000005155.70870.BD/METRICS>
- Goldhaber, D. D., & Brewer, D. J. (1997). Why don’t schools and teachers seem to matter?: Assessing the impact of unobservables on educational productivity. *Journal of Human Resources*, 32(3), 505–523.
<https://doi.org/10.2307/146181>
- Goodhart, C. A. E. (1984). Problems of Monetary Management: The UK Experience. *Monetary Theory and Practice*, 91–121. https://doi.org/10.1007/978-1-349-17295-5_4
- Grohnert, T., Gromotka, L., Gast, I., Delnoij, L., & Beausaert, S. (2024). Effective master’s thesis supervision – A summative framework for research and

- practice. *Educational Research Review*, 42, 100589.
<https://doi.org/10.1016/J.EDUREV.2023.100589>
- Grönroos, C. (1984). A Service Quality Model and its Marketing Implications. *European Journal of Marketing*, 18(4), 36–44. <https://doi.org/10.1108.4784>
- Grönroos, C., & Ojasalo, K. (2004). Service productivity: Towards a conceptualization of the transformation of inputs into economic results in services. *Journal of Business Research*, 57(4), 414–423. [https://doi.org/10.1016/S0148-2963\(02\)00275-8](https://doi.org/10.1016/S0148-2963(02)00275-8)
- Hanushek, E. A. (1979). Conceptual and Empirical Issues in the Estimation of Educational Production Functions. *The Journal of Human Resources*, 14(3), 351. <https://doi.org/10.2307/145575>
- Hanushek, E. A., & Ettema, E. (2017). Defining Productivity in Education: Issues and Illustrations. *American Economist*, 62(2), 165–183. <https://doi.org/10.1177/0569434516688207>
- Hattie, J., & Marsh, H. W. (1996). The relationship between research and teaching: A meta-analysis. *Review of Educational Research*, 66(4), 507–542. <https://doi.org/10.3102/00346543066004507>
- Herberholz, L., & Wigger, B. U. (2021). Efficiency of European universities: A comparison of peers. *Research Policy*, 50(9), 104314. <https://doi.org/10.1016/J.RESPOL.2021.104314>
- Hicks, D., Wouters, P., Waltman, L., De Rijcke, S., & Rafols, I. (2015). Bibliometrics: The Leiden Manifesto for research metrics. *Nature*, 520(7548), 429–431. <https://doi.org/10.1038/520429a>
- Hill, T. P. (1977). ON GOODS AND SERVICES. *Review of Income and Wealth*, 23(4), 315–338. <https://doi.org/10.1111/J.1475-4991.1977.TB00021.X>
- Hopp, W. J., & Spearman, M. S. (2021). The lenses of lean: Visioning the science and practice of efficiency. *Journal of Operations Management*, 67(5), 610–626. <https://doi.org/10.1002/JOOM.1115>
- Hulten, C. R. ., Dean, Edwin., & Harper, M. J. . (2001). *New developments in productivity analysis*. University of Chicago Press.
- Humlum, M. K., & Thorsager, M. (2021). The Importance of Peer Quality for Completion of Higher Education ☆. *Economics of Education Review*, 83, 102120. <https://doi.org/10.1016/j.econedurev.2021.102120>
- Jain, I., & Gulati, R. (2025). Efficiency in Higher Education: A Review of Research From 1977 to 2022. *Higher Education Quarterly*, 79(1). <https://doi.org/10.1111/HEQU.70010>
- Johnes, G., & Johnes, J. (2009). Higher education institutions' costs and efficiency: Taking the decomposition a further step. *Economics of Education Review*, 28(1), 107–113. <https://doi.org/10.1016/J.ECONEDUREV.2008.02.001>
- Johnes, J. (2006). Data envelopment analysis and its application to the measurement of efficiency in higher education. *Economics of Education Review*, 25(3), 273–288. <https://doi.org/10.1016/J.ECONEDUREV.2005.02.005>

- Johnes, J. (2008). Efficiency and Productivity Change in The English Higher Education Sector from 1996/97 to 2004/5*. *The Manchester School*, 76(6), 653–674. <https://doi.org/10.1111/J.1467-9957.2008.01087.X>
- Johnes, J., Portela, M., & Thanassoulis, E. (2017). Efficiency in education. *Journal of the Operational Research Society*, 68(4), 331–338. <https://doi.org/10.1057/S41274-016-0109-Z>
- Johnes, J., & Yu, L. (2008). Measuring the research performance of Chinese higher education institutions using data envelopment analysis. *China Economic Review*, 19(4), 679–696. <https://doi.org/10.1016/J.CHIECO.2008.08.004>
- Jongbloed, B., Vossensteyn, H., van Vught, F., & Westerheijden, D. F. (2018). Transparency in Higher Education: The Emergence of a New Perspective on Higher Education Governance. *European Higher Education Area: The Impact of Past and Future Policies*, 441–454. https://doi.org/10.1007/978-3-319-77407-7_27
- Kalyuga, S., Ayres, P., Chandler, P., & Sweller, J. (2003). The Expertise Reversal Effect. *Educational Psychologist*, 38(1), 23–31. https://doi.org/10.1207/S15326985EP3801_4
- Kao, C. (2009). Efficiency decomposition in network data envelopment analysis: A relational model. *European Journal of Operational Research*, 192(3), 949–962. <https://doi.org/10.1016/J.EJOR.2007.10.008>
- Kendrick, J. W. (1956). *Productivity Trends Capital and Labor*.
- Kong, W. H., & Fu, T. T. (2012). Assessing the performance of business colleges in Taiwan using data envelopment analysis and student based value-added performance indicators. *Omega*, 40(5), 541–549. <https://doi.org/10.1016/J.OMEGA.2011.10.004>
- Kopcsó, D., & Pachamanova, D. (2018). Business value in integrating predictive and prescriptive analytics models. *INFORMS Transactions on Education*, 19(1), 36–42. <https://doi.org/10.1287/ITED.2017.0186CA>
- Lazear, E. (2001). Educational Production. *The Quarterly Journal of Economics*, 116(3), 777–803. <https://doi.org/10.1162/00335530152466232>
- Lee, B. L., & Johnes, J. (2022). Using network DEA to inform policy: The case of the teaching quality of higher education in England. *Higher Education Quarterly*, 76(2), 399–421. <https://doi.org/10.1111/HEQU.12307>
- Lepori, Benedetto., Bonaccorsi, Andrea., Daraio, Alessandro., Daraio, Cinzia., Gunnes, Hebe., Hovdhaugen, Elisabeth., Ploder, Michael., Scannapieco, Monica., & Wagner-Schuster, Daniel. (2016). *Establishing a European tertiary education register : final report*. 87.
- Lewis, M. A., & Brown, A. D. (2012). How different is professional service operations management? *Journal of Operations Management*, 30(1–2), 1–11. <https://doi.org/10.1016/J.JOM.2011.04.002>
- Lillrank, P., & Liukko, M. (2004). Standard, routine and non-routine processes in health care. *International Journal of Health Care Quality Assurance*, 17(1), 39–46. <https://doi.org/10.1108/09526860410515927/FULL/PDF>

- Lovelock, C., & Gummesson, E. (2004). Whither Services Marketing? *Journal of Service Research*, 7(1), 20–41. <https://doi.org/10.1177/1094670504266131>
- Lovelock, C. H. (1983). Classifying Services to Gain Strategic Marketing Insights. *Journal of Marketing*, 47(3), 9–20. <https://doi.org/10.1177/002224298304700303>
- Marginson, S. (2008). Academic Creativity Under New Public Management: Foundations for an Investigation. *Educational Theory*, 58(3), 269–287. <https://doi.org/10.1111/J.1741-5446.2008.00288.X>
- Marsh, H. W. (2007). Do University Teachers Become More Effective With Experience? A Multilevel Growth Model of Students' Evaluations of Teaching Over 13 Years. *Journal of Educational Psychology*, 99(4), 775–790. <https://doi.org/10.1037/0022-0663.99.4.775>
- Marsh, H. W., & Hattie, J. (2002). The relation between research productivity and teaching effectiveness: Complementary, antagonistic, or independent constructs? *Journal of Higher Education*, 73(5), 603–641. <https://doi.org/10.1080/00221546.2002.11777170>
- Massy, W. F., Sullivan, T. A., & Mackie, C. (2013). Improving Measurement of Productivity in Higher Education. *Change: The Magazine of Higher Learning*, 45(1), 15–23. <https://doi.org/10.1080/00091383.2013.749140>
- Meeusen, W., & van den Broeck, J. (1977). Efficiency Estimation from Cobb-Douglas Production Functions with Composed Error. *International Economic Review*, 18(2), 435–444. <https://about.jstor.org/terms>
- Merton, R. K. (1968). The Matthew Effect in Science. *Science*, 159(3810), 56–62. <https://doi.org/10.1126/SCIENCE.159.3810.56>
- Metters, R., & Vargas, V. (2000). A typology of de-coupling strategies in mixed services. *Journal of Operations Management*, 18(6), 663–682. [https://doi.org/10.1016/S0272-6963\(00\)00038-3](https://doi.org/10.1016/S0272-6963(00)00038-3)
- Moed, H. F. (2005). Citation Analysis in Research Evaluation. In *Dordrecht: Springer Netherlands*. Springer-Verlag. <https://doi.org/10.1007/1-4020-3714-7>
- Morris, B., & Johnston, R. (1987). Dealing with Inherent Variability: The Difference Between Manufacturing and Service? *International Journal of Operations & Production Management*, 7(4), 13–22. <https://doi.org/10.1108/EB054796>
- Morrison, H. G., Magennis, S. P., & Carey, L. J. (1995). Performance Indicators and League Tables: A Call for Standards. *Higher Education Quarterly*, 49(2), 128–145. <https://doi.org/10.1111/J.1468-2273.1995.TB01671.X>
- Nelson, P. (1970). Information and Consumer Behavior. *Source: Journal of Political Economy*, 78(2), 311–329.
- O'Donnell, C. J., Rao, D. S. P., & Battese, G. E. (2008). Metafrontier frameworks for the study of firm-level efficiencies and technology ratios. *Empirical Economics*, 34(2), 231–255. <https://doi.org/10.1007/S00181-007-0119-4/METRICS>
- OECD. (2025a). *Education at a Glance 2025 OECD INDICATORS*. <https://doi.org/10.1787/1c0d9c79-en>

- OECD. (2025b). OECD Economic Surveys: Finland 2025. *OECD Economic Surveys: Finland, OECD Economic Surveys: Finland, 2025*.
<https://doi.org/10.1787/985D0555-EN>
- OECD. (2025c). *The Financial Sustainability of Higher Education. Higher Education*.
<https://doi.org/10.1787/F544CCFE-EN>
- Parasuraman, A., Zeithaml, V. A., & Berry, L. L. (1988). SERVQUAL: A Multiple-Item Scale for Measuring Consumer Perceptions of Service Quality. *Journal of Retailing*, 64(1).
- Parteka, A., & Wolszczak-Derlacz, J. (2013). Dynamics of productivity in higher education: Cross-european evidence based on bootstrapped Malmquist indices. *Journal of Productivity Analysis*, 40(1), 67–82.
<https://doi.org/10.1007/S11123-012-0320-0/TABLES/12>
- Pine, J., Victor, B., & Boynton, A. C. (1993). Making Mass Customization Work. *Harvard Business Review*, 71(5).
- Pruvot, E. B., & Estermann, T. (2022). *Allocating core public funding to universities in Europe: state of play & principles*. www.eua.eu
- Räty, T., & Kivistö, * -Jussi. (2006). *Mitattavissa oleva tuottavuus Suomen yliopistoissa*. <https://www.doria.fi/handle/10024/148596>
- Ray, S. C., & Desli, E. (1997). Productivity Growth, Technical Progress, and Efficiency Change in Industrialized Countries: Comment. *The American Economic Review*, 87(5), 1033–1039.
- Robledo, M. A. (2024). Customer transformation in services: conceptualization and research agenda. *Journal of Service Theory and Practice*, 34(3), 420–431.
<https://doi.org/10.1108/JSTP-01-2023-0012>
- Salas-Velasco, M. (2024). Nonparametric efficiency measurement of undergraduate teaching by university size. *Operational Research*, 24(1), 1–29.
<https://doi.org/10.1007/S12351-024-00816-X/TABLES/4>
- Sampson, S. E. (2000). Customer-supplier duality and bidirectional supply chains in service organizations. *International Journal of Service Industry Management*, 11(4), 348–364. <https://doi.org/10.1108/09564230010355377>
- Sampson, S. E. (2012). Visualizing service operations. *Journal of Service Research*, 15(2), 182–198. <https://doi.org/10.1177/1094670511435541/FORMAT/EPUB>
- Sampson, S. E. (2018). Professional Service Jobs: Highly Paid but Subject to Disruption? *Service Science*, 10(4), 457–475.
<https://doi.org/10.1287/SERV.2018.0227>
- Sampson, S. E., & Chase, R. B. (2022). Optimizing Customer Involvement: How Close Should You Be to Your Customers? *California Management Review*, 65(1), 119–146.
https://doi.org/10.1177/00081256221118117/SUPPL_FILE/SJ-PDF-1-CMR-10.1177_00081256221118117.PDF
- Sampson, S. E., & Froehle, C. M. (2006). Foundations and implications of a proposed Unified Services Theory. *Production and Operations Management*, 15(2), 329–343. <https://doi.org/10.1111/J.1937-5956.2006.TB00248.X/FORMAT/EPUB>

- Sampson, S. E., Pires dos Santos, R., Scott Sampson, C. E., Bendoly, E., Fransoo, J., Hoberg, K., Ou, C., & Tenhiälä, A. (2023). Reengineering professional services through automation, remote outsourcing, and task delegation. *Journal of Operations Management*, 69(6), 911–940. <https://doi.org/10.1002/JOOM.1268>
- Sauder, M., & Espeland, W. N. (2009). The Discipline of Rankings: Tight Coupling and Organizational Change. *American Sociological Review*, 74(1), 63–82. <https://doi.org/10.1177/000312240907400104>
- Silvestro, R., Fitzgerald, L., Johnston, R., & Voss, C. (1992). Towards a Classification of Service Processes. *International Journal of Service Industry Management*, 3(3), 62–75. <https://doi.org/10.1108/09564239210015175>
- Simar, L., & Wilson, P. W. (2000). A general methodology for bootstrapping in non-parametric frontier models. *Journal of Applied Statistics*, 27(6), 779–802. <https://doi.org/10.1080/02664760050081951>
- Simar, L., & Wilson, P. W. (2007). Estimation and inference in two-stage, semi-parametric models of production processes. *Journal of Econometrics*, 136(1), 31–64. <https://doi.org/10.1016/J.JECONOM.2005.07.009>
- Solow, R. M. (1957). Technical Change and the Aggregate Production Function. *Source: The Review of Economics and Statistics*, 39(3), 312–320. <https://www.jstor.org/stable/1926047>
- Spence, M. (1978). Job Market Signaling. *Uncertainty in Economics*, 281–306. <https://doi.org/10.1016/B978-0-12-214850-7.50025-5>
- Spring, M., & Araujo, L. (2009). Service, services and products: rethinking operations strategy. *International Journal of Operations & Production Management*, 29(5), 444–467. <https://doi.org/10.1108/01443570910953586>
- Star, C., & McDonald, J. (2007). Embedding Successful Pedagogical Practices: Assessment Strategies for a Large, Diverse, First Year Student Cohort. *International Journal of Pedagogies and Learning*, 3(2), 18–30. <https://doi.org/10.5172/IJPL.3.2.18>
- Steenbergen-Hu, S., Makel, M. C., & Olszewski-Kubilius, P. (2016). What One Hundred Years of Research Says About the Effects of Ability Grouping and Acceleration on K–12 Students’ Academic Achievement: Findings of Two Second-Order Meta-Analyses. *Review of Educational Research*, 86(4), 849–899. <https://doi.org/10.3102/0034654316675417>
- Stephan, P. (2015). *How economics shapes science*. Harvard University Press. <https://books.google.com/books?hl=fi&lr=&id=b5svE-AAAQBAJ&oi=fnd&pg=PA2004&dq=stephan+2015+economics+shapes+science&ots=JwteVolcAX&sig=auii-X4Kt83DK-2f-C8DOVVP958>
- Stewart, D. M., & Chase, R. B. (1999). The Impact of Human Error on Delivering Service Quality. *Production and Operations Management*, 8(3), 240–263. <https://doi.org/10.1111/J.1937-5956.1999.TB00307.X>
- Susskind, R., & Susskind, D. (2017). The Future of the Professions: How Technology Will Transform the Work of Human Experts. *Journal of Nursing Regulation*, 8(2), 52. [https://doi.org/10.1016/s2155-8256\(17\)30099-6](https://doi.org/10.1016/s2155-8256(17)30099-6)

- Syversen, C. (2011). What Determines Productivity? *Journal of Economic Literature*, 49(2), 326–365. <https://doi.org/10.1257/JEL.49.2.326>
- Tinto, V. (1975). Dropout from Higher Education: A Theoretical Synthesis of Recent Research. *Review of Educational Research*, 45(1), 89–125. <https://doi.org/10.3102/00346543045001089/ASSET/E7ACE700-E536-4B2B-8530-126F2C24977D/ASSETS/00346543045001089.FP.PNG>
- Triplett, J. E. ., & Bosworth, Barry. (2004). *Productivity in the U.S. services sector : new sources of economic growth*. 401.
- Uttl, B., White, C. A., & Gonzalez, D. W. (2017). Meta-analysis of faculty’s teaching effectiveness: Student evaluation of teaching ratings and student learning are not related. *Studies in Educational Evaluation*, 54, 22–42. <https://doi.org/10.1016/J.STUEDUC.2016.08.007>
- Vaarma, M., & Li, H. (2024). Predicting student dropouts with machine learning: An empirical study in Finnish higher education. *Technology in Society*, 76, 102474. <https://doi.org/10.1016/J.TECHSOC.2024.102474>
- Vakkuri, J., & Meklin, P. (1998). Tulosmittaus ja »vastiketta rahalle»-ajattelutapa - näkökulmia mittaus teoreettisiin ongelmiin ja riskeihin. In *HALLINNON TUTKIMUS* (Vol. 2).
- Vargo, S. L., & Lusch, R. F. (2004). Evolving to a New Dominant Logic for Marketing. *Journal of Marketing*, 68(1), 1–17. <https://doi.org/10.1509/JMKG.68.1.1.24036/FORMAT/EPUB>
- Waltman, L. (2016). A review of the literature on citation impact indicators. *Journal of Informetrics*, 10(2), 365–391. <https://doi.org/10.1016/J.JOI.2016.02.007>
- Wemmerlöv, U. (1990). A Taxonomy for Service Processes and its Implications for System Design. *International Journal of Service Industry Management*, 1(3), 20–40. <https://doi.org/10.1108/09564239010002126>
- Winston, G. C. (1999). Subsidies, Hierarchy and Peers: The Awkward Economics of Higher Education. *Journal of Economic Perspectives*, 13(1), 13–36. <https://doi.org/10.1257/JEP.13.1.13>
- Winston, G., & Zimmerman, D. (2004). Peer Effects in Higher Education. *College Choices: The Economics of Where to Go, When to Go, and How to Pay for It*, (University of Chicago Press), 395–424.
- Worthington, A. C. (2001). An Empirical Survey of Frontier Efficiency Measurement Techniques in Education. *Education Economics*, 9(3), 245–268. <https://doi.org/10.1080/09645290110086126>
- Worthington, A. C., & Lee, B. L. (2008). Efficiency, technology and productivity change in Australian universities, 1998–2003. *Economics of Education Review*, 27(3), 285–298. <https://doi.org/10.1016/J.ECONEDUREV.2006.09.012>
- WTO. (2023). The future of trade lies in services: key trends. *Harnessing Services Trade for Sustainable Growth*, 17–49. <https://doi.org/10.30875/9789287075406C005>

- Yorke, M. (1999). Assuring quality and standards in globalised higher education. *Quality Assurance in Education*, 7(1), 14–24.
<https://doi.org/10.1108/09684889910252496>
- Zeithaml, V. A. (1981). How Consumer Evaluation Processes Differ Between Goods and Services. *Marketing of Services*, 9(1), 25–32.
- Zeithaml, V. A. (1988). Consumer Perceptions of Price, Quality, and Value: A Means-End Model and Synthesis of Evidence. *Journal of Marketing*, 52(3), 2–22. <https://doi.org/10.1177/002224298805200302>
- Zeithaml, V. A., Parasuraman, A., & Berry, L. L. (1985). Problems and Strategies in Services Marketing. *Journal of Marketing*, 49(2), 33–46.
<https://doi.org/10.1177/002224298504900203>

Appendix

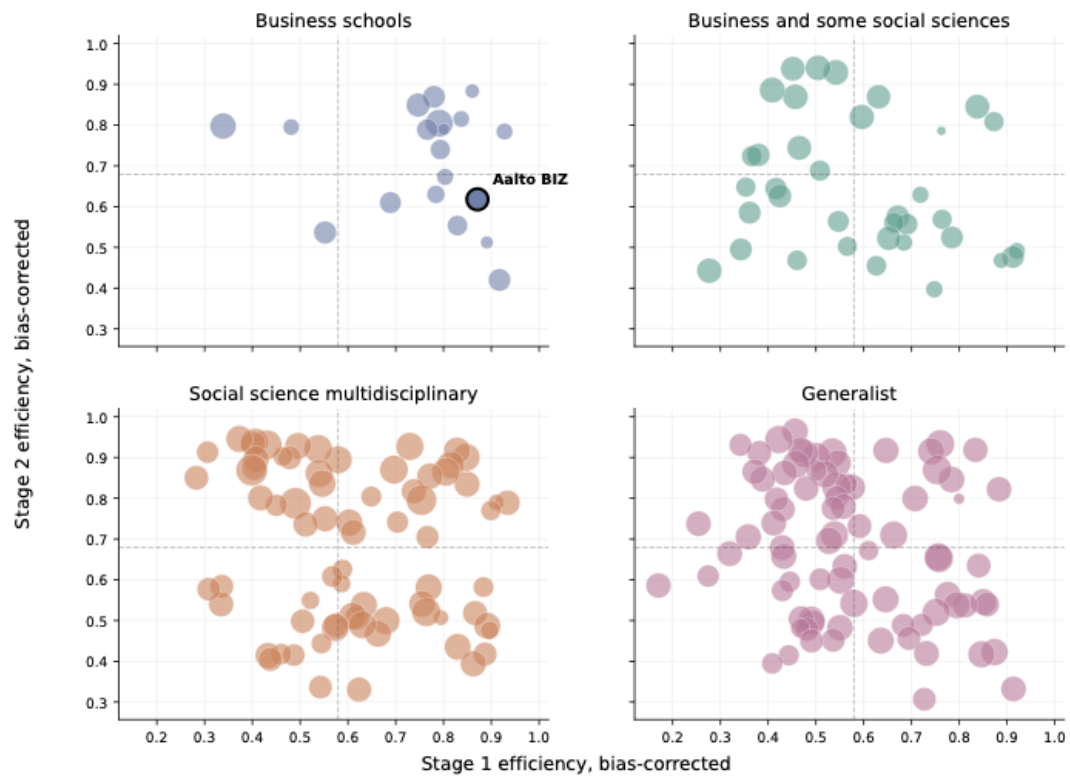


Figure 10. Efficiency scatter by cluster.

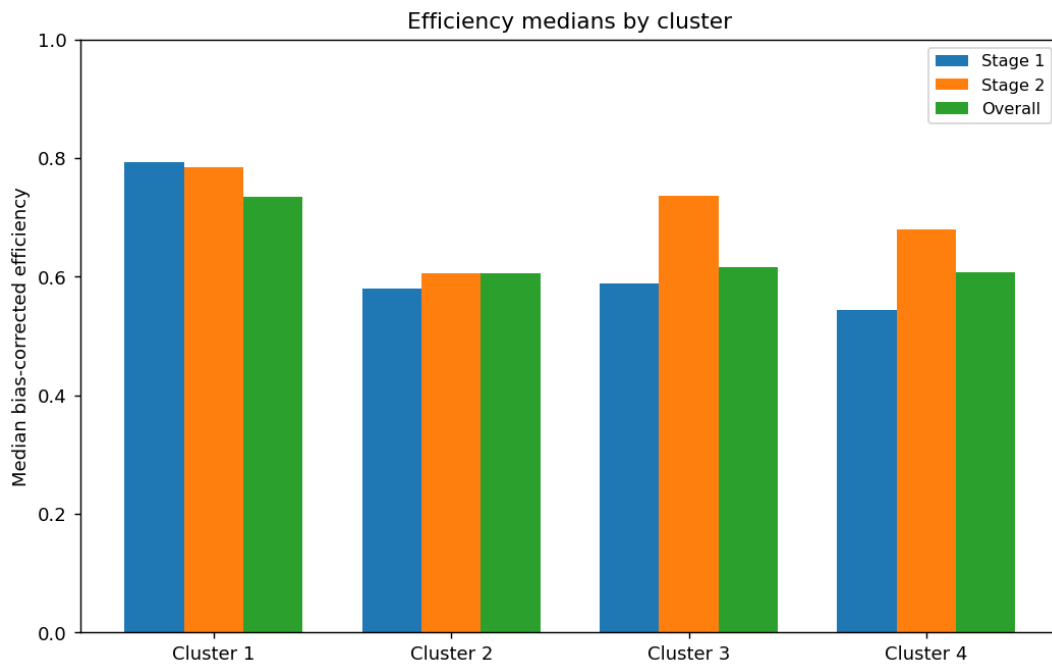


Figure 11. Efficiency medians by stage and cluster.

Cluster 1: Business schools
Cluster 2: Business and social science focus
Cluster 3: Broad social science focus
Cluster 4: Broad generalist

Table 7. Bias-corrected DEA efficiency scores for business schools.

Institution	Stage 1	Stage 2	Overall
EBS University of Business and Law	0.86	0.88	0.87
Norwegian School of Economics	0.93	0.78	0.85
Stockholm School of Economics	0.84	0.82	0.83
Copenhagen Business School	0.78	0.87	0.82
Erasmus University Rotterdam	0.79	0.80	0.80
Tilburg University	0.75	0.85	0.80
Zeppelin University Friedrichshafen	0.80	0.79	0.79
BI Norwegian Business School	0.77	0.79	0.78
Frankfurt School of Finance & Management-HfB	0.79	0.74	0.77
University of Economics in Bratislava	0.80	0.67	0.74
Aalto BIZ	0.87	0.62	0.73
ISCTE - University Institute of Lisbon	0.78	0.63	0.70
University of Economics, Prague	0.83	0.55	0.68
Hanken School of Economics	0.89	0.51	0.68
University of Mannheim	0.69	0.61	0.65
Vienna University of Economics and Business	0.92	0.42	0.62
WHU – Otto Beisheim School of Management	0.48	0.80	0.62
University of Sankt Gallen	0.55	0.54	0.54
London School of Economics and Political Science	0.34	0.80	0.52

Note: Stage 1, Stage 2, and overall scores are bias-corrected DEA efficiency estimates. Overall efficiency is calculated as the geometric mean of the two stage scores. Institutions are sorted by overall efficiency in descending order.

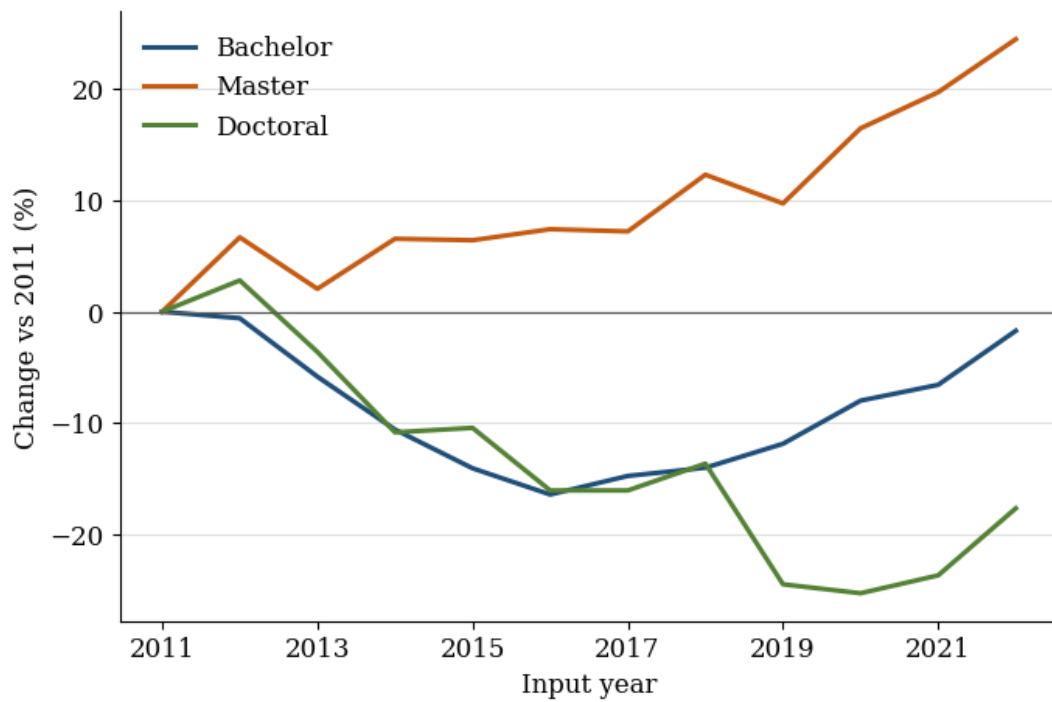


Figure 12. Aalto BIZ yearly students by study level.

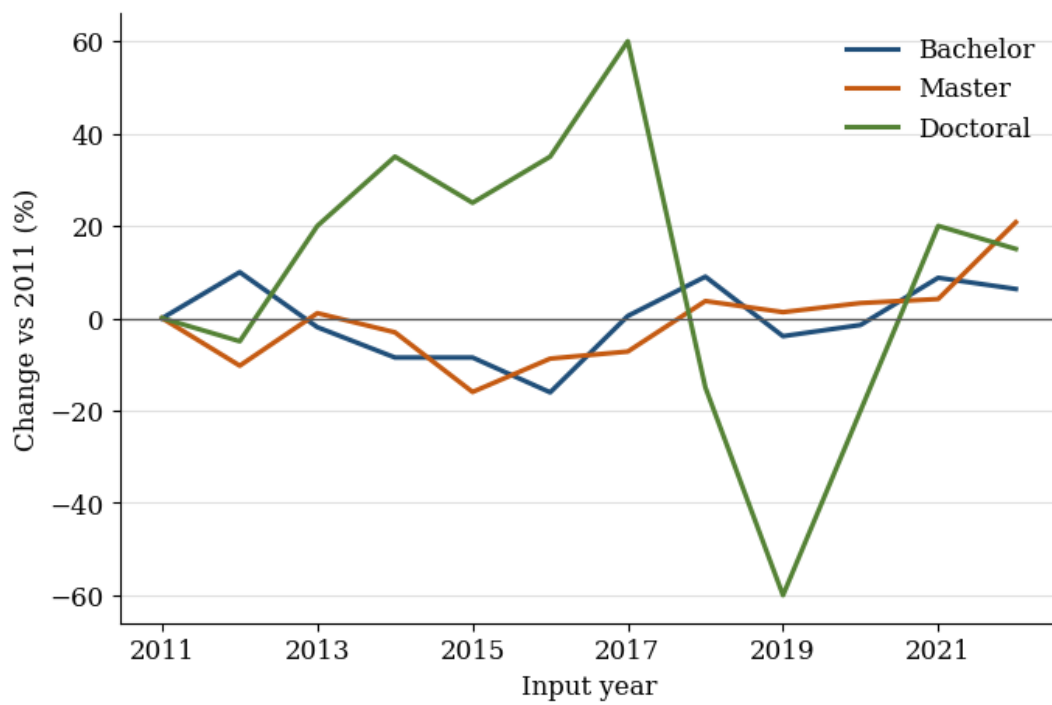


Figure 13. Aalto BIZ yearly graduates by study level.

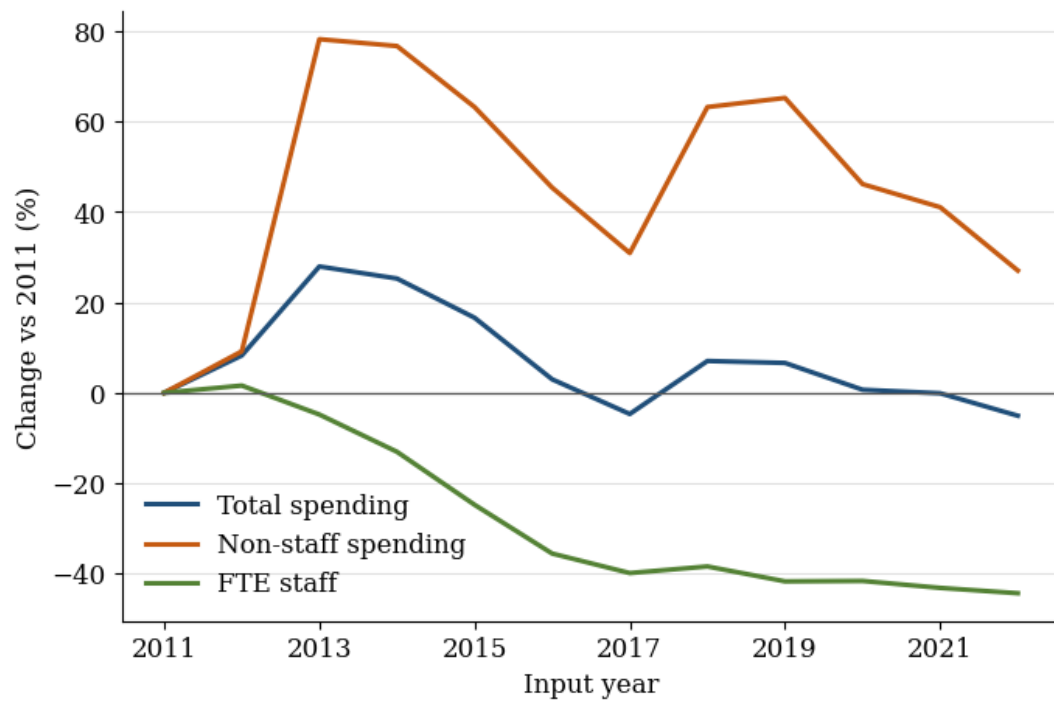


Figure 14. Aalto BIZ yearly resource use.

Table 8. Stationary input-output ratios across the entire sample.

Institution	Students/ FTE	Spending/ student €	Spending/ graduate €	Graduates/ 100 students	Master's stu- dent share %	Master's gradu- ate share %	Doctoral stu- dent share %
Aalto BIZ	11.9	10,025	41,056	24.4	48.4	52.2	5.8
Cluster 1 me- dian	8.9	10,875	41,056	32.1	42.6	54.5	3.3
Cluster 2 me- dian	6.5	14,732	49,692	26.6	29.0	46.1	6.2
Cluster 3 me- dian	4.9	17,759	75,901	22.8	26.7	40.9	6.8
Cluster 4 me- dian	4.3	18,863	85,434	23.0	30.8	45.6	8.4
Resource-size peers average	10.1	14,231	45,202	30.2	40.8	52.1	5.4
Resource- composition peers average	10.4	14,290	46,781	29.3	38.3	50.9	4.8
Technology peers average	11.0	12,738	41,794	29.3	38.0	49.6	4.5